

Four Choices Behind One Number: Reporting the Incremental Value of a Recorded Field

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Abstract

Papers, business cases and tool evaluations report what a recorded field is worth as a single number. That number is a function of four choices which are almost never stated: the *baseline* of already-recorded fields the comparison is allowed to contain, the *metric*, the *operating point*, and the *population* of the register the field reads from. We write the quantity as $V(f \mid B, m, \theta)$ and propose that a feature-value claim be reported as a surface over those arguments rather than as a point.

We measure the practice rather than asserting it. A pre-registered audit of 600 papers, enumerated from a public index and screened by a rule fixed before the first paper was read, yields 54 that report the incremental value of a named feature; 30 were read and 20 confirmed in scope. Two in five state their baseline and half argue their metric, so the strong complaint is false and we withdraw it. **Not one of the twenty reports the increment across a range of operating points, and not one across a range of register populations** — the two axes on which our own answer moves most.

The demonstration is a configuration management database (CMDB) on a public event log of 45,455 incidents from a bank’s IT operation, predicting at intake whether an incident will be reassigned. Each choice moves the answer by more than the effect the previous version of this paper reported. Admitting one further field the organisation already records — which group logged the incident — takes the item’s value from +0.183 AUC to +0.103. Admitting a second, which a file we had not obtained now shows is written by the service desk before the incident exists, takes it to +0.001 [−0.002, +0.003]: the headline does not survive our own admissibility criterion once the evi-

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dence to apply it is in hand. Under six defensible instruments on the same scores the reduction runs 43.7% to 60.3%, and the AUC figure we published is the smallest of them; under net benefit it runs 4.7% to 134.1% across the operating range, and the item is resolvably harmful in a band above the base rate. At half population the item is worth +0.082 if the register’s long tail is missing and +0.064 if its core is.

The whole surface is then run on all three logs whose reduction is resolvable — three organisations, three information systems, two domains — and on every one of them the baseline choice alone moves the reduction by more than the reduction itself. A simulation whose true reduction is known by exhaustive enumeration of a 960-cell space puts the estimator’s bias at most 0.0044 and its bootstrap coverage between 0.850 and 0.975.

We pre-registered a generic protocol and applied it to 22 public event logs across seven domains. It admits 13; the reduction is resolvably positive on three. The generality claim we registered is therefore falsified, and we report the negative, the exclusions and the condition that does govern: on most logs the entity is worth nothing to the intake baseline, so there is no value for a free field to absorb. That condition was then registered as a prediction, with its parameter frozen, and tested on two logs this project had never opened; it produced two correct predictions, both negative, and therefore no test of its positive half, which we report as the limitation it is.

The reporting standard ships as `fieldvalue`, a package that computes the surface and refuses by design to emit a single number. Re-deriving this paper’s headline through it agrees with our own pipeline on 20 of 20 quantities to 2.2×10^{-16} , and found an error in this paper’s own prose on its first run.

Twelve errors of our own are reported as results rather than edited away, and the paper reports the defect *classes* they fall into rather than the incidents alone. Three are claims whose every individual number was correct and whose asserted relation between them was not; the most recent was found by an independent implementation of our own estimator.

Keywords: incremental value, baseline specification, evaluation metrics, predictive process monitoring, configuration management database, variable importance, decision curve analysis, pre-registration

1. Introduction

A business case for a data programme is a comparison, and the second arm of that comparison — performance *without* the data — is a choice the analyst makes, usually silently. So is the metric the comparison is stated in, so is the operating point at which that metric is read, and so is the assumed completeness of the register being bought. Each of those choices is routinely left implicit, and the result is reported as one number.

We write the quantity such a claim actually names as

$$V(f \mid B, m, \theta), \tag{1}$$

the incremental value of a feature f over a baseline feature set B , under metric m , at operating point θ ; and for a feature that is a lookup into a register we treat that register’s population as a fourth choice, one that acts on f rather than on the comparison. The claim of this paper is not that V depends on those arguments — that is a theorem, and Section 3 says whose — but that on a real decision, with every choice held to a defensible range, the dependence is larger than the effect being reported. We demonstrate it by taking our own previously published headline apart along each argument in turn. One of those demonstrations removes the headline.

The demonstration runs on a configuration management database (CMDB): the register of items an IT estate contains, bought to make operational analytics possible, evaluated here on a named task — predicting at intake whether an incident will be reassigned — on a public log of 45,455 incidents from a bank’s IT operation.

This paper reports four things.

1. **The practice failure is real, and measured.** We pre-registered an audit — the frame, the retrieval rule, the inclusion rule, five codes and the reliability design, committed before a single paper was screened — and ran it over 600 papers enumerated from a public index. Of the 54 that report the incremental value of a named feature, we read 30 and confirm 20 as in scope. Two in five state their baseline and half argue their metric, so the strong form of the complaint is *false* and we withdraw it. What is true is narrower and sharper: 45.0% report the increment at more than one level of the metric and 30.0% at more than one baseline, and **not one of the twenty reports it across a range of operating points, or across a range of register populations**

(Section 4). Those are the two axes on which this paper’s own answer moves most.

2. **What can happen, before what does: three propositions.** R is metric-unbounded — there are score distributions on which two defensible metrics give 0 and any r up to 0.998; R is invariant under a strictly monotone transformation of the scores *if and only if* the metric is rank-based, which says exactly which instruments can see calibration; and no one of the four choices determines another, for ten of the twelve ordered pairs, with the two exceptions named and explained. Each is constructed and executed as a test (Section 3.3, Appendix Appendix A).
3. **Every choice moves the answer by more than the effect — on three organisations, and against an answer that is known.** Admitting one further field the organisation already records takes the item’s value from +0.183 to +0.103 [+0.094, +0.113] AUC; admitting a second, the knowledge-article reference, takes it to +0.001 [−0.002, +0.003]. On identical scores six instruments put the reduction between 43.7% and 60.3%; across the operating range it runs 4.7% to 134.1%. The whole surface is then run on the three logs whose reduction is resolvable, and on every one of them the baseline choice alone moves R by more than R itself (Section 10). A simulation whose true reduction is known by exhaustive enumeration puts the estimator’s bias at most 0.0044 and its interval coverage between 0.850 and 0.975 (Section 11).
4. **A falsified generality claim, a registered prediction, a standard you can run, and twelve corrections.** The registered claim that the effect is a property of process event logs rather than of ITSM data is *falsified* on 22 public logs. The condition that does govern was registered as a prediction and tested on two logs this project had never opened — which produced two correct predictions, both negative, and therefore no test of the rule’s positive half, and we say so (Section 12.3). The reporting standard ships as `fieldvalue`, a package that computes the surface and refuses to emit a single number; re-deriving this paper’s headline through it agrees with our own pipeline on 20 of 20 quantities and found the twelfth correction (Sections 13 and 13.1).

What kind of contribution this is.. We report no deployed system. This is a retrospective study on two public benchmarks, offered as an evaluation

practice. Section 8 states what the choice costs in units closer to those a service desk operates in, and Section Appendix C states which *layer* of configuration data carries the value, which is the question a buyer actually faces.

2. Background

Prediction on incident event logs.. Our task sits at the boundary of predictive process monitoring, surveyed and benchmarked across real-world logs [24, 25] on data of the kind process mining formalises [1]. We predict at $t = 0$ from creation-time attributes with no prefix, which is tabular classification on a process log rather than prefix-based monitoring in the sense those benchmarks evaluate; we note that Teinemaa et al. [24] explicitly excluded the 2013 and 2014 BPI Challenge logs from their benchmark for lacking a static/dynamic attribute split. The BPI Challenge logs are the standard public instances [8, 22]; comparable exports exist elsewhere [3]. Such logs carry documented quality hazards [23]. One hazard bears directly on us and we state it as our own measurement rather than attributing it: values recorded once per record can be replicated onto every event of that record, which makes constancy across a case uninformative about when the value became known. Establishing what a resource stamp on an event denotes is the business of organisational mining [2], which is what Section 5 does from first principles for the field our result turns on. The state of the queue a case arrives into — the inter-case perspective — carries real predictive content in this setting [21], and Section 6 admits four free creation-time congestion features to the baseline for that reason.

The CMDB and what justifies it.. A CMDB records configuration items and their relationships, and is the asset on which the configuration-management practice of ITIL and its successors rests. Survey work on ITSM framework adoption [15] finds that realised benefits grow with implementation maturity, but measures them as practitioner-reported outcomes rather than as the analytic value of the configuration data itself.

What is and is not already measured.. Prior peer-reviewed work uses the affected configuration item as one input among many without isolating its value. Schad et al. [20] predict ticket reassignment — our own target — with a relational graph convolutional network on a public ServiceNow log, with the configuration item among the incident-node features. Sarnovský and

Surma [19], working on the same Rabobank log we use, report ROC AUC and rank CI type, CI subtype and service component among the most significant attributes. Kapel et al. [11] predict change-induced incidents at a large bank and rank CI Name third by SHAP importance, behind two free-text fields. None of the three performs a controlled with-and-without comparison attributing a change in a fixed metric to configuration-item identity against a stated baseline, which is what we do; but the reader should note that the question is not untouched, and that in the one study which ranks a free-text field against the configuration item, free text wins.

The methods a CMDB is bought to enable.. Surveys of AIOps for failure management [30, 17, 16] catalogue triage, failure-prediction and root-cause methods that take configuration data as input. Incident triage in particular — deciding which team owns an incident — has been treated as a learning problem on operational logs at industrial scale [5], as has failure prediction across a cloud estate [14].

What the comparison is measured against.. The value of a feature *given* a baseline is what algorithm-agnostic variable-importance frameworks formalise; Williamson et al. [28] develop this for R^2 -type predictiveness and Williamson et al. [29] extend it to AUC, which is our metric. Our overlap is the redundancy that additive importance measures share credit for [7]. Section 3 says what this paper adds to those frameworks and what it does not.

The estimand itself — the change in a fixed metric when one variable joins an existing model — has been argued over in biostatistics for two decades, and that literature reached our conclusion first. Cook [6] shows that an increment in AUC is a poor measure of a variable’s incremental value and depends heavily on the model it is added to. Vickers and Elkin [26] supply the principled replacement for the capacity-limited framing we used in earlier versions and withdraw in Section 8. We claim no priority over these; we report that the failure mode they describe is, on operational data, large enough to change a procurement decision.

Within machine learning, Rendle et al. [18] and Ferrari Dacrema et al. [9] show insufficiently tuned baselines accounting for reported gains in recommender systems; benchmark conclusions are sensitive to variation that is seldom reported [4], and documentation of the choices producing a result is thin [10]. Where the choice concerns which fields may enter, it shades

into leakage [13, 12], whose formal treatments accommodate admission rules as constraints set by the problem definition rather than derived from it. In predictive process monitoring, Weytjens and De Weerd [27] show that avoiding a different leak — case prefixes shared between training and test under naive temporal splitting — needs explicit construction; we adopt their strict temporal splitting and, at their prompting, examine the extract boundary in Section 5. We report a field-admission failure mode for a deployment decision rather than a leaderboard, where it is not a modelling detail but the number the business case turns on.

3. This Is Not Just Conditional Variable Importance

3.1. The estimand

Equation 1 names $V(f \mid B, m, \theta) = m(B \cup \{f\}, \theta) - m(B, \theta)$. For two nested baselines $B_0 \subset B_1$ we report the *admissibility reduction*

$$R(f \mid B_0, B_1, m, \theta) = 1 - \frac{V(f \mid B_1, m, \theta)}{V(f \mid B_0, m, \theta)}, \quad (2)$$

the share of a feature’s apparent value already carried by fields the organisation records for nothing. θ is absent for metrics that integrate over the operating range; Section 7 says which those are. We report R only where $V(f \mid B_0, m, \theta)$ is positive in every bootstrap draw. A ratio whose denominator crosses zero is not a quantity, and this paper has previously printed one that did.

Three properties of R should be stated before it is used, because each is a limit a reader would otherwise have to infer.

First, R is metric-relative *by construction*. It divides two increments of m , and an increment of AUC has no natural zero of the kind a ratio needs, so “43.7% of the value is absorbed” under AUC and “60.3%” under average precision are not one quantity measured twice. We do not claim R is comparable across m ; Section 7 reports the spread across m as the finding.

Second, every interval in this paper comes from a paired bootstrap that resamples test rows and holds the fitted models fixed. It therefore quantifies test-set sampling error and not model-fitting variability, and is narrower than a full resampling would give. The substitute is the design-space sweep this paper already carries — split point, target threshold, cleaning cutoff and three estimator families — across which the AUC reduction runs 36.1% to

48.3%. A reader should treat that range, not a bootstrap interval, as the honest width.

Third, four is a claim about how many choices are usually left *implicit*, not about how many exist. The estimator family is a fifth and the cohort a sixth; the literature already treats the estimator as a choice, and this paper varies it over three families and reports the range. The four below are the ones a feature-value claim routinely fixes without saying so.

Four independent choices therefore sit inside any single-number claim: which already-recorded fields B contains, which m , which θ , and — for a feature that is a lookup into a register — how completely that register is populated. Sections 6, 7, 8 and 9 vary one at a time on one dataset, holding the cohort, the split, the estimator and the seed fixed throughout, so that any movement is attributable to the choice and to nothing else. Section 13 states what we propose be reported in place of a point.

3.2. *What is and is not new here*

The objection we expect first is that measured importance is baseline-relative, that Williamson et al. [29] and Covert et al. [7] formalise exactly that, and that a study reporting it on one log is reporting a known fact with a new number attached. The objection is correct about the formal content and we concede it in those terms. Four things are added here that no formal framework supplies.

A magnitude on a named decision.. Those frameworks establish that a feature’s importance is defined only relative to a baseline. They do not say how large the dependence is for any particular feature on any particular decision, and a procurement decision turns on the size. On this task the answer is that the choice of baseline moves the measured value of an entire data programme by roughly half, and that half is the difference between an investment case and none.

The right estimand is not an average over baselines.. A Shapley-style attribution averages a feature’s contribution over subsets of the others, which is the correct thing to compute when the question is how to apportion credit within a fixed model. It is the wrong thing to compute here. A business case needs a feature’s value against *the baseline the organisation will actually have* — the fields already in its ticketing tool — not an average over baselines it

will never be in. We therefore report a ladder over named, admissible baselines rather than a single attribution, and Section 5 states the admissibility criterion before applying it.

The overlapping field is free.. The interesting case is not that some baseline choice changes the answer; any correlated feature does that. It is that the field which absorbs half the measured value is one the organisation *already records at no cost*, sitting in the same export, requiring no programme and no budget line. That is what makes the analyst’s choice invisible rather than merely arbitrary: nothing in the data flags the omission, and the omitted field has no price tag to prompt a question about it.

The metric argument is not in those frameworks at all.. Williamson et al. [29] and Covert et al. [7] are stated for a fixed predictiveness measure. Nothing in them says that two defensible measures applied to the *same scores* disagree about how much of a feature’s value a second feature absorbs, still less by how much. That is an empirical question, Section 7 answers it here, and the answer is that on this decision the metric choice moves the reduction by more than the free field does.

Where it appears and where it does not, measured rather than assumed.. Baseline-relativity is a theorem; whether a particular pair of fields overlaps in a particular estate is an empirical question with no general answer. We pre-registered a protocol and applied it to 22 public logs (Section 12) rather than generalising from one. The answer is mostly negative and is reported as such.

What we do not claim.. We do not claim a new estimator, a new importance measure, or priority over Cook [6], who stated the underlying point for AUC in 2007. Nor do we claim any of the four choices is illegitimate: each is correct about the question it asks. The contribution is that those are different questions, that a single number answers one of them without saying which, and that on this decision the spread between them is larger than the effect.

3.3. Three propositions about what can and cannot happen

That V depends on its arguments is a theorem and not ours. What is not in the frameworks above is *how far* the dependence can go, *which* instruments can see *which* choice, and whether knowing the effect of one choice constrains the effect of another. Three small results, each with a constructive proof that

a reader can execute (`scripts/r41_propositions.py`; the constructions are in Appendix Appendix A). They are small and we say so: their job is to move this paper’s contribution from “measured some magnitudes” to “established what can and cannot happen, and then measured what does”.

Proposition 1 (metric-unboundedness).. For a target $r \in [0, 1)$ there exist score distributions, a baseline B_0 , a feature f and two defensible metrics under which the admissibility reduction R equals r and 0 respectively.

Construction. Let every model’s score take one of two values, so each is described by the counts (a, b) of positives and negatives in the high block. With P positives, $N = \rho P$ negatives, $\alpha = a/P$, $\beta = b/N$ and $\pi = 1/(1 + \rho)$, two identities hold exactly: $\text{AUC} - \frac{1}{2} = (\alpha - \beta)/2$, and $\text{AP} - \pi = \alpha(\alpha/(\alpha + \beta\rho) - \pi)$. Take both baselines to be the fully tied model. Fixing $\alpha - \beta = d$ on both legs makes the AUC increments identical, so R under AUC is *exactly* zero; sliding α along $[d, 1]$ moves the AP increment monotonically. Executed over ten targets, R under ROC AUC is zero to 4.4×10^{-16} — floating-point noise, not approximation — while R under average precision hits every target from 0.05 to 0.99 to within 1.8×10^{-4} , **on identical scores**. The declared family reaches 0.998; that bound is reported rather than assumed away, and it widens with ρ .

This is the analytic form of what Section 7 measures: 43.7% under AUC against 60.3% under average precision, on scores that do not move.

Proposition 2 (the rank-invariance boundary).. R is invariant under every strictly increasing transformation of a model’s scores **if and only if** the metric is rank-based.

Proof. $(\Leftarrow)$ A strictly increasing φ preserves the weak ordering, so a rank-based m is unchanged, so every V is unchanged, so R is. $(\Rightarrow)$ If m is not rank-based there are s, s' inducing the same weak ordering with $m(s) \neq m(s')$; equality of the weak ordering makes $s_i \mapsto s'_i$ well defined and strictly increasing on the distinct values of s , and any such map extends to a strictly increasing φ on the real line. Applying it to one leg changes that leg’s V and hence R . (QED)

Witness. On a synthetic ladder, applying one strictly increasing φ to the naive leg’s scores moves R by exactly 0.000000 under ROC AUC and average precision and by between 1.645 and 4.255 under Brier skill, Nagelkerke R^2 and net benefit at two operating points.

The proposition converts Section 7’s recalibration result from an observation into a statement about *which instruments can see calibration at all*:

recalibrating the models moves the proper-score reductions and leaves the rank-based ones at exactly zero, and no implementation detail can change that.

Proposition 3 (non-redundancy)... No one of the four choices determines another: for an ordered pair of axes (X, Y) there exist two configurations agreeing on how much X moves the answer and differing on how much Y does.

Construction, and its limit. Over a declared family of 18 synthetic estates — varying how tightly the opening field is coupled to the entity, how much of the entity’s signal it carries, and how strong the entity is — we search all twelve ordered pairs. **Ten of the twelve are constructible** at the declared tolerances. The two that are not are (metric, threshold) and (population, threshold), and the reason is stated rather than hidden: the threshold axis’s spread across this family is an order of magnitude larger than the others, so the “differ by a quarter of the axis’s own range” criterion demands a gap of 3.396 and the best pairs found reach 1.907 and 2.913. That is a property of the tolerance and of the threshold axis’s scale, not evidence that those pairs are redundant, and we report the two failures rather than retuning until twelve appear.

4. How the Field Reports Feature Value Today

Everything above is an argument that four choices set the answer. A referee is entitled to reply that the argument is a known critique — Cook on AUC increments, Hand on metric incoherence, Vickers and Elkin on operating points, Williamson and Covert on baseline-relativity — and that a critique everybody already accepts needs no paper. The reply to that is a measurement: a critique everybody accepts and nobody follows is not a known critique but a documented practice failure, and this section documents one.

4.1. What was registered, and when

AUDIT-PROTOCOL.md fixes the frame, the retrieval rule, the inclusion rule, five codes and the reliability design. It was committed *before a single paper was screened*, in a commit that adds no `results/r40_*.csv` and no downloaded full text; `git log -stat` shows it arriving alone. Section 1 of that document states, in advance, the outcome that would refute this paper’s premise: a majority of included papers stating their baseline *and* a majority

justifying their metric. That outcome did not arrive, but something close enough to it did that the paper’s wording had to change, and §4.4 says so.

The frame.. Two frames, unioned and deduplicated on DOI, enumerated through the OpenAlex API so that a reader can re-run the query rather than trust a description of it. *Frame A*: research articles published 2019–2026, open access, in *Information Systems*, *Decision Support Systems*, *Information & Management* and *Empirical Software Engineering*, passing a declared topical pre-filter on title and abstract. *Frame B*: every open-access work 2019–2026 citing either OpenAlex record of Teinemaa et al.’s outcome-oriented predictive process monitoring benchmark [24].

The conference venues, and why they are not in the frame.. The plan this round executed named BPM, ICPM and CAiSE. They are not machine-enumerable in OpenAlex and we do not pretend otherwise: measured on 2026-08-22, before any screening, the BPM source record carries 69 works for 2019–2026 and *none* flagged open access, the CAiSE record carries one, and a search for ICPM returns an unrelated serial. Most papers from those venues are indexed under the umbrella LNCS source, which cannot be filtered to a conference. A frame that cannot be reproduced is not a frame. Those venues are covered through Frame B instead, and the venue mix of the included set is reported below whatever it turns out to be.

4.2. The funnel

The two frames yield 607 works, 604 after deduplication, and the registered cap of 600 is applied by simple random sample with seed 20260819. Full text is retrieved for 369 of the 600 — 61.5% — and this is the stage that loses the most, for reasons nobody can show are related to the codes and which §16 declines to assume away. Of the 369, 192 report no quantitative performance metric and 123 carry no comparison of performance with and without a named feature. **The included set is 54 papers across 15 venues.**

The screening rule is a pattern set fixed in `r40_audit.py` before the first paper was read, and every included paper’s record carries the sentence that admitted it, so the inclusion decision is as auditable as the codes. Reading the first run’s output found three patterns that did not implement the rule written above them — the “with/without” class admitted “Without basic knowledge over the data domain”, and three of the five codes matched sentences

about something other than the model’s evaluation. Amendment 1 to the protocol records all three, and *both* codings are reported: the registered screen admits 117 papers, the amended screen 54, and `results/r40_proportions.csv` carries the proportions under each.

4.3. *The codes, and what a mechanical coder is worth*

Five binary items per included paper, each with a mechanical rule, each carrying a verbatim quote and the PDF page it was found on: `B_stated` (is the baseline’s feature set enumerable from the paper?), `B_justified` (is the exclusion of an already-available field argued?), `M_justified` (is the metric argued for, not merely named?), `Theta_stated` (is an operating point or a cost ratio stated, or is the metric explicitly said to integrate over them?), and `Range_reported` (is the *increment* reported at more than one level of any of the four choices?).

A mechanical coder is not a reader, and this paper’s primary numbers are not the mechanical ones. Thirty included papers were drawn with the registered seed and coded by reading a deterministic extract of each — every sentence carrying a metric and a number, and every sentence carrying baseline, feature, threshold, cost or metric-justification vocabulary — written to `data/audit/adjudication/` so that the evidence a code was assigned from is as public as the code. Two things came out of that reading and both are reported rather than repaired.

First, **the screen’s precision is 66.7%**: of the thirty papers the amended screen admitted, a read confirms 20 as reporting a quantitative comparison of predictive performance with and without a named feature. The other ten compare a resampling technique, a preprocessing pipeline, a set of method components or, in one case, human reading behaviour.

Second, **the mechanical coder under-detects badly**. Against the read it missed 39 `yes` codes and asserted 9 a read does not support; on `Range_reported` it found none where a read finds 11 of 20. The mechanical proportions are therefore reported as a *lower bound*, in those words, and the adjudicated proportions on $n = 20$ are what Table 1 prints first.

4.4. *The result, and what it does to this paper’s premise*

The strong claim is false and comes out.. We expected to find that a large majority of papers state none of the four choices. They do not. Two of five enumerate their baseline, half argue their metric, and three in five state an

Table 1: How the field reports the incremental value of a field. Adjudicated by reading, on the 20 papers a read confirms are in scope; the mechanical coder’s proportions on all 54 included papers are printed beside them and are a lower bound, not an estimate. Intervals are Wilson binomial 95%.

code	adjudicated, $n = 20$		mechanical, $n = 54$	
	yes	share	yes	share
B_stated	8	40.0% [21.8, 61.4]	4	7.4% [2.9, 17.6]
B_justified	5	25.0% [11.1, 46.9]	6	11.1% [5.1, 22.2]
M_justified	10	50.0% [29.9, 70.1]	10	18.5% [10.3, 30.9]
Theta_stated	12	60.0% [38.6, 78.2]	14	25.9% [16.1, 39.0]
Range_reported	11	55.0% [34.2, 74.2]	0	0.0% [0.0, 6.7]

operating point or say explicitly that their metric integrates over one. A field that does that is not ignoring the critique.

What survives is narrower and, we think, sharper.. Reporting the increment at more than one level of *some* choice is common. It is almost never more than one choice, and it is almost always the same two:

- over the **metric**: 9 of 20, 45.0% [25.8, 65.8];
- over the **baseline**: 6 of 20, 30.0% [14.5, 51.9];
- over the **operating point**: **0 of 20**, 0.0% [0.0, 16.2];
- over the **register’s population**: **0 of 20**, 0.0% [0.0, 16.2];
- over **two or more** of the four at once: 4 of 20, 20.0% [8.0, 41.7].

Not one paper in the subsample reports what a feature is worth across a range of operating points, and not one across a range of register populations. Those are exactly the two axes on which this paper’s own headline moves most: Section 8 puts the reduction between 4.7% and 134.1% across the operating range, and Section 9 moves it by a quarter of its own size across the population curve. The practice failure this paper documents is not that the field ignores its own critiques. It is that the field varies the two choices that are cheap to vary and does not vary the two that decide the answer.

Reliability, with one author.. A single author coded every paper, by machine, and then read thirty. That is not inter-rater reliability and this paper does not call it that. What is reported instead: every code ships with its quote and page, so the coding is auditable rather than trusted; a second, independently written implementation of the same five rules agrees with the first on a random 20% of the included set with Cohen’s κ between 0.256 and 0.501, which is *poor*, and is printed here because it is the honest measure of how much the operationalisation matters; and the seven-day blind re-code by the author is written into `submission/OWNER-ACTIONS.md` with the sample already drawn, because it needs elapsed time and a human, and an agent has neither.

4.5. *What this costs a reader, worked on a public dataset*

The consequence is easiest to see away from event logs. UCI Adult (doi:10.24432/C5XW20) predicts whether income exceeds \$50k. Suppose an analyst reports what `occupation` is worth: it is the expensive field, the one a survey has to ask about and code, where age, sex, race and country come free with the sampling frame and education comes with the enrolment record. Running the surface with the package this paper ships:

- against the fields that come free with the frame, admitting `occupation` on top of them absorbs $R = +0.749$ of what it looked worth against nothing;
- against those plus education and hours worked, $R = +0.200$.

Same data, same feature, same metric, same operating point. **The baseline alone moves the answer by three quarters of its own size.** We name no defect in any paper in doing this, and the point is not that anyone was careless: the practice is universal, this paper’s own earlier versions included, and Section 17 is the evidence for that.

5. Data and Task

5.1. *The primary organisation*

We use the incident records of the BPI Challenge 2014 log [8] from Rabobank Group ICT, recorded in HP Service Manager, with the activity log shipped alongside. The configuration-item column we use is `CI Name (aff)`, the item affected at the time the incident was recorded, not the closure-time `CI Name (CBy)`.

Cleaning. Of 46,809 rows, 203 carry no incident identifier and one an unparseable timestamp. A further 1,150 opened before 2013-10-01 are left-censored — already open when the extract began — and are reassigned at 81.2% against 40.0% for the incidents we keep. Removing them leaves **45,455** incidents to 2014-03-31; a temporal 70/30 split gives 31,818 training and 13,637 test, positive rates 0.413 and 0.372. The cutoff is not taken from the outcome: monthly volume rises from 857 incidents in September 2013 to 8,606 in October. The extract is right-censored too — incidents opened near its end cannot have long lives — so we re-measured on cohorts truncated up to a month early; the shrinkage moves between 42% and 45%, so the boundary does not drive it. The affected-item field is 100% populated across 2,929 items, 2,554 seen in training. That population rate is a property of the extract and of a tool configuration that copies the item onto the incident from the originating interaction record; it is not evidence about CMDB accuracy in general, and Section 16 returns to the point.

Task and estimator. Predict at creation whether an incident will later be reassigned. We call the target reassignment rather than misrouting throughout: it fires on 37.2% of test incidents, which is routine handling and not error, and Section 8 reports the ladder at stricter thresholds. Median handling time rises from 1.59 hours for incidents never reassigned to 47.53 hours for those reassigned five or more times; we use this to motivate the task, not as a recoverable saving.

The intake block is four fields carried on the incident form: Category, Impact, Urgency and Priority. It is effectively three. **Priority** is a deterministic function of (**Impact**, **Urgency**) on this log — 19 occupied cells, none carrying more than one Priority, across all 46,809 rows — so it adds no information, and dropping it moves intake AUC from 0.562 to 0.564. Sarnovský and Surma [19] reported this redundancy on this same log in 2018. We keep the column because it is what an analyst building the naive arm would include, and we disclose it because a paper arguing that baselines are under-specified should not ship an under-specified baseline in silence. The four fields together take only 23 distinct combinations in training and 23 in test, which is a fact Section 8 needs and which no choice of estimator can change.

The primary estimator is one-hot logistic regression, chosen because it is bin-free: scikit-learn’s **LogisticRegression** with L2 penalty, inverse regularisation strength $C = 1.0$, the **lbfgs** solver and a cap of 3000 iterations, with **OneHotEncoder** set to ignore unseen levels. The penalty is held fixed

across every arm; tuning it per arm on an inner split of training data moves the headline reduction from 43.7% to 43.2%. Two further estimator specifications are reported in Section 6. Encoders and rankings are fit on training data only; intervals are 2,000-resample paired bootstraps, except Table F.7, whose counts are re-ranked in each draw and use 400. Monte-Carlo nulls use 15 to 100 draws, the count set by cost and recorded in the code that produces each. Every sampling script is seeded from a single constant, so the reported figures are reproducible exactly.

The free field, and what it is not.. The activity log records, on the `Open` event of every incident, an assignment group: 49 in training, none missing. An earlier draft of this paper called it the queue the service desk routed the incident to. That was an assertion, not a finding, and it is wrong. Three checks say so. The `Open` rows carry 50 distinct groups where the `Assignment` rows carry 218, and a routing destination cannot be less diverse than the teams that then do the work. One group accounts for 67.0% of `Open` rows but only 18.4% of all activity rows. And the `Open` group matches the group on the incident’s first `Assignment` activity for just 15.1% of the 37,577 incidents that have one. The field is the group that *logged* the incident: one central desk for two thirds of tickets, a long tail for the rest. We do not characterise that tail — its openers match the first `Assignment` group only 21.8% of the time and appear anywhere in the later work rows only 59.1%, so calling them teams opening their own work would be another assertion of the kind we are correcting. It is also concentrated and unstable: the largest group holds 62.1% of training incidents, and the number of live groups falls from 49 to 32 between our two halves, while in the test half it logs 78.6% of incidents.

The free field is not a restatement of the target.. A reader may reasonably suspect the opposite of what Section Appendix D argues: that a field naming a *group* cannot help but encode a target defined as a *change of group*, making the whole ladder circular. That reading makes a directional prediction, and the data contradicts it. The central desk logs most tickets and does a small minority of the work, so on the circular reading it should be the one handing work on, and should reassign more than everyone else. It reassigns *less*: 0.309 against 0.603, a difference of -0.294 $[-0.315, -0.275]$. What the field separates is a low-reassignment intake channel from a high-reassignment one. Nor is it a strong predictor in its own right: the per-group outcome rate, applied as a lookup with no model, scores 0.642, and the one-bit central-desk contrast 0.606. A field that restated the target would score near the ceiling.

Why not the real routing decision.. The obvious repair is not available. The first **Assignment** group occurs strictly after **Open** on every incident that has one, a median of 46 minutes later, and 7,878 of 45,455 never have one. It fails the criterion below, and substituting it would introduce the leakage that criterion prevents.

Admissibility, and mutation.. A field on the **Open** row is admitted only if it is a per-event observation, tested by whether it varies across an incident’s activity rows; the opening group varies for 92.56% of cohort incidents and matches the last-observed group for only 21.35%. Because the incident file is a closed-record snapshot we also report how often each field is edited after creation: Urgency on 1,107 incidents (2.44%), Impact on 1,084 (2.38%), the affected item on 159 (0.35%). Incidents whose Impact was edited are reassigned at 0.66 and those whose Urgency was edited at 0.67, against 0.39 for the rest, so this is not neutral; Section 6 reports the headline with those fields dropped and with the edited incidents removed. The activity log carries no Category or Priority change type, so for those two fields the snapshot cannot be audited at all; our disclosure covers the fields for which an audit exists, not every field we use.

A second criterion governs the other direction, and Section Appendix C needs it: a field may join the CMDB-free baseline only if the organisation would hold it *without* the CMDB. That excludes every field which is a function of item identity, because such a field exists only where the item register does.

The second organisation.. A second public ITSM log, BPI Challenge 2013 incidents from Volvo IT Belgium, carries the same construction on a different estate and a different tool; its cohort, its target and what does and does not transfer are in Appendix Appendix G, and its full surface is in Section 10.

6. The Admissibility Effect

Against four intake fields the affected item looks decisive. Adding the opening group cuts its measured value by 44%. The gains are roughly 28 and 17 standard deviations outside a null in which the incident-to-item association is shuffled, preserving column count and marginal distribution while destroying the signal — we round because re-drawing the same null at a different draw count moves the first figure by about 0.7 — a comparison that matters because adding item identity adds 2,554 indicator columns, which is

Organisation	Baseline	AUC	+ item id.	gain [95% CI]
Rabobank	intake only	0.562	0.746	+0.183 [+0.172, +0.195]
(BPIC 2014)	+ opening group	0.644	0.748	+0.103 [+0.094, +0.113]
Volvo IT	intake only	—	—	+0.238
(BPIC 2013)	+ opening group	—	—	+0.092

Table 2: The value of knowing the affected configuration item, with and without a field the organisation already records, on two organisations. The Volvo rungs use a coarser item identifier and a more tightly coupled free field; see Section Appendix G.

not free under a fixed penalty. These z figures pool the null’s Monte-Carlo dispersion with the gain’s own bootstrap standard error; dividing by the null alone, as an earlier draft did, would report 51 and 33. Because these nulls use as few as 15 draws, a two-digit z is an extrapolation from a dispersion estimate and not a tail probability; the load-bearing statement is that no null draw comes near either gain.

It is not a split artifact: across six split points the two gains range +0.175 to +0.194 and +0.091 to +0.119 (Figure 2b), and across split point, target threshold and cleaning cutoff the second ranges +0.067 to +0.130 — wider than any bootstrap interval, which conditions on all of those. Nor does the disclosed field editing explain it: reducing the intake block to Category alone gives +0.106, and restricting to the 44,227 never-edited incidents gives +0.107.

The reduction, over the whole design space.. The reduction rather than either gain is what we argue transfers, so it needs the same disclosure. Across split points it runs 38.7% to 48.3%; across the three target thresholds 43.6% to 48.3%; across the three estimator specifications 42.7% to 47.1%; and across five cleaning cutoffs from no censoring removal at all to a two-month truncation, 36.1% to 44.0%. Over the whole design space it runs **36.1% to 48.3%**. The bootstrap interval of [40, 48] we report at the published setting conditions on every one of those knobs, and an earlier version of this paper quoted it without saying so while asserting in its title that the reduction was nearly a half. At the low end of the design space it is closer to a third. What survives across the space is that the reduction is large and its sign never changes.

It is not congestion.. A reader from the process-mining community will ask whether the affected item is standing in for load, since the inter-case perspec-

tive is known to carry real content on this kind of log [21] and neither of our baselines looks at the state of the queue an incident arrives into. Four features are free at creation and are computed from the open and close timestamps of *other* incidents, every one of which is in the past at the moment of creation: the open backlog, arrivals in the preceding hour, hour of day and day of week. Continuous features are cut at training deciles, so the estimator is unchanged. Admitted to the baseline on the same criterion as the opening group, they move the item’s measured value from +0.103 to +0.100, and the reduction from 43.7% to 45.7% [41, 50]. Alone, without the intake block or the group, the same four features score 0.497 — at this horizon, on this log, creation-time congestion carries essentially nothing. The item is not a load proxy.

It replicates on a second organisation.. On Volvo IT the same ladder gives +0.238 against intake and +0.092 once the opening group is admitted, a reduction of 61.3% [54, 68]; at the stricter two-change threshold, +0.249 and +0.139, a reduction of 43.9% [31, 55]. Neither interval includes zero. Across six temporal split points the first runs 60.3% to 67.8% and the second 40.8% to 53.8%. The qualitative claim — that admitting one field the organisation already records removes a large fraction of the item’s measured value — holds on a second organisation, a second tool and a second country. The magnitude does not transfer, and Section Appendix G gives a structural reason to expect the Volvo figure to be the larger of the two.

It is not an artifact of the estimator.. The dimensionality null rules out the regularisation burden of 2,554 sparse columns, but a reader may want the effect shown where that burden cannot arise. We re-represent item identity as a single cross-fitted target-encoded column — fitted on training only, out-of-fold, so no row sees its own outcome — which makes the item one continuous feature and makes histogram boosting usable on a variable that otherwise collapses 2,554 items into 137 bins. Across one-hot logistic regression, logistic regression on the encoded item, and boosting on it, the first rung ranges +0.173 to +0.184, the second +0.091 to +0.104, and the reduction 42% to 48%. Encoding a *shuffled* item column under the same pipeline returns -0.0001 ± 0.0016 and $+0.0042 \pm 0.0020$ on the group-aware rung. The first collapses to zero; the second does not, and it is eleven standard errors from zero, so it is a real systematic effect rather than noise and it must be bounded on both rungs and not one. Run on the intake rung as well, the same control

returns $+0.0002 \pm 0.0015$ and -0.0036 ± 0.0025 . Subtracting each rung’s own residual from that rung’s gain moves the reduction from 42.7% to 42.6% under the encoded logistic model and from 47.1% to 50.5% under boosting. The correction is therefore worth at most 3.4 percentage points and, where it bites, it makes the reduction larger.

Three target definitions.. Because reassignment is common, we repeat the ladder at stricter thresholds. Requiring two or more reassignments (21.8% of incidents) gives +0.131 and +0.068; three or more (10.6%) gives +0.151 and +0.080. The magnitude is threshold-dependent and not monotone, but the interval on each +group rung excludes zero (we compute intervals for that rung only), the ordering never reverses, and the reduction stays between 43% and 49%.

Does it survive a change of task?. A gap on one target may be a property of it, so we repeated the ladder on two further outcomes within the primary organisation. *Reopening* is close to an independent failure mode: 2,096 incidents, correlating with reassignment at +0.14. *Long handling* — handle time above the training third quartile — correlates at +0.40, so it is corroboration only.

On reopening, item identity is worth +0.083 against the intake block and +0.055 once the group is admitted, a reduction of 33%; on long handling, +0.118 and +0.078, a reduction of 34%. Every rung sits outside its matched-dimension null. But the reduction is what we argue transfers, so it needs its own interval, and paired bootstraps give $[40, 48]$ on reassignment, $[28, 38]$ on long handling and $[-1, 60]$ on reopening. On the one target we argue is near-independent the reduction is *not* resolvably different from zero ($P \leq 0$ is 0.03), and long handling, correlated at +0.40 with the primary target, cannot carry a replication alone. Reopening is rare and its gains stand only 5.5 and 4.2 pooled standard deviations from their nulls.

Cross-target replication within one organisation therefore fails to resolve, while cross-organisation replication succeeds. We take the second as the stronger evidence, because the objection it answers — that the effect is a property of one estate — is the one a reader should press hardest, and because both Volvo intervals exclude zero. The reduction is largest on the target the opening group is most directly about, which is consistent with Section Appendix D rather than predicted by it: the mechanism says nothing about how the reduction should order across targets.

instrument	$V(f \mid B_0)$	$V(f \mid B_1)$	reduction
ROC AUC	+0.183	+0.103	43.7% [40.1, 47.1]
Average precision	+0.226	+0.090	60.3% [57.2, 63.4]
Brier skill	+0.168	+0.070	58.4% [54.4, 62.4]
Nagelkerke R^2	+0.223	+0.097	56.6% [52.8, 60.3]
Net benefit, $\theta = 0.325$	+0.093	+0.087	6.3%
Net benefit, $\theta = 0.500$	+0.084	-0.016	119.2%

Table 3: The same reduction under four instruments that integrate over the operating range and two readings of one that does not. B_0 is the four intake fields; B_1 adds the opening group. Intervals are paired bootstrap, 2,000 draws, one resampled index set per draw shared by all four models. The net-benefit rows carry no interval on the ratio because the whole point of the row pair is that the ratio is not one number.

7. The Metric Axis

Section 6 varied B and held m at ROC AUC. This section holds B and the scores fixed and varies m . Same cohort, same split, same four fitted models, same seed: the only thing that changes is what is computed from the scores.

7.1. Seven instruments, six of them independent

Table 3 reports the reduction of Equation 2 under every instrument we can defend for this question. Detection at a fixed review capacity is kept in the table although Section Appendix F withdrew it as a headline, because a matrix of instruments is the right place to show what a tie-degenerate instrument does to a ratio.

Expected cost at a stated cost ratio is *not* a seventh axis. At cost ratio r the Bayes threshold is $1/(1+r)$, so the net-benefit weight $\theta/(1-\theta)$ is exactly $1/r$ and the cost avoided per case is $r \cdot \text{NB}(1/(1+r))$ identically. The two therefore give the same reduction; we verified it over ten contrasts, where the two agree to machine precision, and report six instruments rather than seven.

The number we published is the smallest of them.. Across the four instruments that integrate over the operating range the reduction runs 43.7% to 60.3%, and ROC AUC sits at the bottom of that range. The previous version of this paper reported the AUC figure and treated the more principled

instruments as a threat to it; on this data they are not, and a referee reading only Section Appendix F would have drawn the opposite conclusion. This is the second correction in this project’s history that goes against the direction its errors usually take, and we record it as such rather than quietly banking it.

Why the instruments disagree.. Three mechanisms account for the disagreement — a diffuse increment that AUC sums and net benefit reads one point of, a baseline that is treat-all at eleven of the sixteen grid points, and two tie conventions the rank statistics do not share. They are measured in Appendix Appendix E, and none of them makes one instrument the right one.

8. What the Difference Buys: the Operating-Point Axis

An AUC difference is not a quantity a service desk can act on, so earlier versions of this paper stated the same comparison as detection at a fixed review capacity: rank the test incidents by predicted risk, take the top 5%, and count how many reassignment-bound incidents each model surfaces. That framing is withdrawn here, and the reason is correction seven in Section 17.

Why the capacity framing was withdrawn.. The previous version stated this axis as detection at a fixed review capacity and reported a factor of 4.3. That framing is withdrawn: at a 5% capacity 93.1% of what the naive baseline nominates comes from one tied block, and reordering rows inside it — which changes nothing any model knows — moves the naive arm from -26 to $+608$. Appendix Appendix F is the full withdrawal, and it is correction seven.

Net benefit, which needs no capacity.. Decision curve analysis states the axis without a capacity: at an operating point θ a desk is willing to review $\theta/(1 - \theta)$ non-events per event, and net benefit counts detections against reviews at that exchange rate. The full grid, its intervals, the treat-all and treat-none references and the band in which admitting item identity to a group-aware model makes the desk *worse off* are in Appendix Appendix K. Two numbers from it are load-bearing here and are stated rather than deferred: the increment is resolvably negative in a contiguous band above the base rate, reaching -21.1 $[-27.7, -14.4]$ per thousand at $\theta = 0.525$, and across that band the group-aware item model acts on 17% to 37% of arrivals.

8.1. *The range across the operating range, corrected*

Across the 28 grid points at which the naive increment’s own interval excludes zero, the reduction runs from 4.7% at $\theta = 0.175$ to 134.1% at $\theta = 0.575$. The three remaining grid points are reported in `results/r30_nb_grid.csv` and excluded from the range because a ratio whose denominator is not resolvable positive is not a quantity.

An earlier version of this sentence gave that range as 6.3% to 119.2%. Those are the values at $\theta = 0.325$ and $\theta = 0.500$, two of the five thresholds Table 3 happens to name, and describing them as the range *across the operating range* named an extremum that was not the extremum. It is correction twelve (Appendix Appendix B), it was found by the independent code path of Section 13.1, and it is the third instance of a class in which every literal is correct and the word joining them is not.

8.2. *What the operational statement does not say*

The threshold ladder of Section 6 is the operational statement of the same robustness, at the three target definitions reported there. None of this says a surfaced incident is a corrected one, and net benefit does not claim otherwise: it counts detections against reviews at a stated exchange rate, and what a detection is worth to a particular desk is a quantity this log cannot supply.

9. The Population Axis

Our item field is 100% populated. Real registers are not, and every practitioner who has read a draft of this paper has said so. That is a measurement, not a caveat, so we make it one: we mask the item column to a declared population level and re-measure the whole ladder, leaving both baselines untouched, so the curve is the value of a partially populated register rather than the value of a smaller dataset. A masked item becomes an explicit empty level rather than a dropped row, because a desk with a half-empty register still has the incidents.

How an estate empties turns out to matter more than how empty it is, so we report three regimes: items masked independently at random; the least frequent items dropped whole, which is how a real estate degrades because discovery and manual curation cover the big central things first; and the most frequent dropped whole, which is not realistic and is reported as the other end of the range. At half population the group-aware increment is +0.082 when the long tail is missing, +0.067 at random, and +0.064 when the core is;

the reduction at the same three points is 39.3%, 36.1% and 14.2%. Over the whole surface the reduction runs -0.6% to 47.1% , and its interval excludes zero at 19 of the 19 points measured. Figure 3 draws it.

The practical reading is that a buyer who prices a half-populated register from the random curve will be wrong in whichever direction their estate actually degrades, and by roughly a factor of three in the reduction. The usual caveat — “our figure is an upper bound because real data is worse” — is not supported on this axis either: at full population the measurement sits at the top of its own curve, so better discovery tooling cannot raise it.

10. The Four Axes, on Three Organisations

Everything to this point varies the four choices on one log, and a referee is entitled to say that “each choice moves the answer by more than the effect” is then an $n = 1$ claim. The corpus of Section 12 finds three logs on which the reduction is resolvable: BPI Challenge 2014 (a Dutch bank’s IT operation, HP Service Manager), BPI Challenge 2013 incidents (Volvo IT Belgium, VINST) and BPI Challenge 2019 (a multinational coatings manufacturer’s procurement process, SAP). Three organisations, three information systems, two domains. `r44_axes_multilog.py` runs the whole surface on all three through one code path: a four-rung baseline ladder giving six ordered nested pairs, four scalar instruments, the 31-point net benefit grid, and four register-population levels under two degradation regimes.

Table 4: The range of the range. For each log, the reduction R at the reference cell, and the interval R occupies as each choice alone is varied with the other three held at the reference. The threshold column is restricted to the grid points where the naive net-benefit increment is at least a tenth of its maximum, because a ratio whose denominator is positive but tiny is not a quantity; the unrestricted range is in `results/r44_spread.csv` and is wider.

log	R	baseline	metric	operating point	population
BPIC 2014	0.350	[0.010, 0.575]	[0.080, 0.381]	[−1.728, 0.694]	[0.228, 0.406]
BPIC 2013 incidents	0.378	[0.356, 0.823]	[0.147, 0.484]	[−0.420, 0.872]	[−0.856, 0.437]
BPIC 2019	0.471	[0.190, 0.844]	[0.471, 0.676]	[−7.502, 1.570]	[0.462, 0.548]

What replicates.. On every one of the three logs, **the baseline choice alone moves R by more than R itself**: the spread is 0.565, 0.467 and 0.654 against reference values of 0.350, 0.378 and 0.471. The metric axis moves it

by 0.301, 0.337 and 0.205 — smaller than the baseline axis on all three, and still between four and nine tenths of the effect being reported. This is the paper’s central claim and it is no longer an $n = 1$ claim.

What does not.. The population axis does *not* behave the same way. It moves R by 0.179 on BPI Challenge 2014, by 0.085 on BPI Challenge 2019, and by 1.293 on BPI Challenge 2013 incidents, where degrading the register to a quarter populated drives R negative. That is a fifteen-fold difference across three logs, and the honest reading is that how much the register’s completeness matters is a property of the estate rather than of the method. Section 9’s curve is a measurement on one estate and the paper does not generalise it.

And the reference cell is a choice too.. The three reference values 0.350, 0.378 and 0.471 look like a replication. They are three points picked from three surfaces whose cells span $[0.009, 0.845]$ once the baseline alone is varied. That agreement is worth exactly as much as the agreement of any three unstated choices, which is the paper’s own argument turned on the paper’s own numbers, and it is why Table 4 prints the intervals beside the points rather than the points alone.

11. The Estimator Against an Answer That Is Known

Every number to this point is estimated and none is checked against a truth, because on a real log there is no truth to check against. A referee who believes the estimator is biased — a ratio of two differences of two fitted AUCs is not an obviously well-behaved thing — has so far had nothing to read. This section builds a world where the answer is known by construction and asks the paper’s own estimator for it.

The construction.. Four discrete variables on a finite cell space: an intake block b with 4 levels, a free opening field g with 6, a high-cost entity f with 40, and a binary outcome whose log-odds are $e_b[b] + (1 - \rho) e_f[f] + \rho e_g[g] + c$. The overlap ρ is the share of the entity’s effect on the outcome that the opening field also carries; g is a many-to-one coarsening of f corrupted at a fixed rate, which is what a resource stamp and a register look like in a real log. The space is 960 cells, so every conditional — and therefore the Bayes-optimal score given any feature set, and therefore the population AUC of that score with ties taking half credit — is computed by **exhaustive enumeration** rather than by sampling. That gives $R^*(\rho)$ exactly.

What the truth looks like.. R^* rises monotonically with the overlap, from 0.133 at $\rho = 0$ — where g still absorbs something, because it is a coarsening of f and predicts on its own — to exactly 1.000 at $\rho = 1$, where $V(f \mid B_0 \cup \{g\})$ is zero to machine precision because g carries everything f has. At $\rho = 0.4$, $R^* = 0.308$; at $\rho = 0.6$, $R^* = 0.600$. The estate that produces the paper’s own headline sits between those two, which is worth stating because it means the measured reduction on BPI Challenge 2014 is what a world with roughly half the entity’s signal in the opening field would produce.

Bias and coverage.. Forty independent replicates per overlap level at $n = 60,000$, each run through the paper’s estimator — one-hot, logistic, 70/30 split by row order, paired bootstrap over test rows with one index set per draw shared by all four models.

Table 5: The estimator against a known answer, 40 replicates at $n = 60,000$ per row. Coverage is the share of replicates whose paired bootstrap 95% interval contains R^* .

overlap ρ	R^*	mean $\hat{R}$	bias	s.d.	coverage
0.0	0.1329	0.1290	−0.0039	0.0112	0.900
0.2	0.1559	0.1514	−0.0044	0.0096	0.975
0.4	0.3078	0.3065	−0.0013	0.0136	0.925
0.6	0.6000	0.6008	+0.0008	0.0139	0.975
0.8	0.8766	0.8787	+0.0021	0.0094	0.975
1.0	1.0000	1.0027	+0.0027	0.0028	0.850

The largest absolute bias is 0.0044 — a third of one standard deviation of the estimator itself — and it shrinks with the sample: the mean absolute bias is 0.0217 at $n = 5,000$ and 0.0025 at $n = 60,000$. Coverage runs 0.850 to 0.975 against a nominal 0.95, mean 0.933. **The estimator this paper uses is very nearly unbiased for the quantity this paper reports, and its interval is close to nominal, on a world where the quantity is known.**

The one place coverage is poor, and why.. At $\rho = 1$ the true reduction is exactly 1, which is the boundary of the parameter space: the estimator’s sampling distribution is squeezed against it, $\hat{R}$ sits +0.0027 above on average, and coverage falls to 0.850. That is the behaviour any interval has at a boundary and it is reported rather than smoothed. It matters for reading

Section 14, whose measured reduction against the intake baseline is 99.7%: an interval near 1 should be read as weaker evidence than one in the middle of the range.

And the surface behaves as the propositions require.. On the same worlds, varying the four choices moves R away from R^* by amounts that differ by axis and by world: the baseline axis by up to 0.606, the threshold axis by up to 4.014, the metric axis by up to 0.187 and the population axis by up to 0.054. The ordering — baseline and operating point large, metric and population smaller — is the same ordering the three real logs produce in Table 4, on data whose generating process nobody chose to make agree with them.

What this does and does not establish.. It validates the *estimator*, not the *estimand*: a world built from a logistic model and fitted with a logistic model is a friendly world, and the bias reported here is a lower bound on the bias against a misspecified truth. It is also entirely era-independent, which answers part — and only part — of the objection that the newest log in the corpus is from 2019.

12. The Corpus: Where the Effect Appears, and Where It Does Not

Everything above is one log. This section asks whether the admissibility effect is a property of process event logs in general or of ITSM data in particular, under a protocol registered before any of it was run.

12.1. Pre-registration

PROTOCOL.md, supplied as supplementary material and committed to the repository in a commit that adds no result file, fixes in advance: the two targets, both reported for every log; the rules that assign every attribute to exactly one of the high-cost entity f , the free per-event field g , the intake block B_0 , a layer, or excluded; six exclusion codes; the estimator, split, seed and instruments; the log-level discriminators, computed before any model is fitted; and the outcome that would falsify the claim. Eight amendments were made after running the role assignment, which reads no outcome, and before fitting any model; each is recorded in that file with its reason, and every one is the registered text failing to implement its own stated intent. The check that they stop there is that after them the generic rules assign the primary

log exactly the roles this paper assigns by hand — f the configuration item, g the opening group, B_0 the four intake fields, and the CI type and subtype as layers — although nothing in the rules names that log or those fields.

The corpus, and what the rules exclude.. The registered rules admit 13 of the 22 logs that parse and exclude nine by code. The exclusions are a finding rather than an accident and are set out in Appendix Appendix H: in BPI Challenge 2012 and every BPI Challenge 2020 sub-log *every case is opened by the same actor*, so there is no free opening field and the question this paper asks cannot arise.

12.2. The result, which is mostly negative

Nineteen log-target pairs survive the exclusion rules. Figure 4 shows every one where a reduction exists to be measured. On 10 of the 19 the entity is not resolvably worth anything over the intake block at all, so there is nothing for a free field to absorb and R is undefined rather than small. On the remaining nine the reduction’s own interval excludes zero on four, drawn from three logs: the primary log on both targets, BPI Challenge 2013’s incident log on the duration target (37.8% [12.8, 62.8]), and BPI Challenge 2019’s procurement log on the duration target (47.1% [40.8, 53.4] on 251,734 traces).

Eight of the 13 admitted logs are outside ITSM, and the reduction is resolvable on one of them. `PROTOCOL.md` §8 states that the claim is falsified if the effect appears on the two published ITSM logs and not on a majority of the admitted non-ITSM logs. It is falsified. We report that rather than reframing the claim to fit, and the paper’s scope is bounded accordingly: what generalises is the *estimand* and the practice of reporting it as a surface, not the magnitude of any particular reduction.

What “falsified” does and does not mean here.. A referee is entitled to object that we did not fail to find the effect so much as fail to have a test: on 10 of the 19 pairs the reduction has no denominator, so it could not have been detected had it been there. The objection is correct and the two statements are separated rather than conflated. The claim registered in `PROTOCOL.md` §8 is falsified *by its own registered criterion*, which we report because pre-registration is worth nothing if the verdict is renegotiated once the answer is known. Beside it: on the nine pairs where a reduction exists to be measured, four are resolvably positive; on the other ten there was nothing to measure. A narrower claim — that where a high-cost entity predicts the outcome at

all, a free opening stamp absorbs a large share of what it is credited with — is neither refuted by this corpus nor asserted by it.

Which target carries this.. Both registered targets are reported for every log, and the four resolvable results are carried by the *duration* target, which is the standard outcome in this literature. The handover target is a generalisation of this paper’s own task rather than a standard one, and Section 12’s validity check below is why we do not lean on it.

Two things the protocol does not do, stated rather than left to be found.. It admits no inter-case features, although Section 6 shows four free creation-time congestion features move the primary log’s reduction from 43.7% to 45.7%; that is one log’s evidence and the corpus does not extend it. And its **NO_HEADROOM** rule reads the target’s prevalence on the full log, which includes the test half. Recomputing that rule on the training half only changes **no** admission decision on any of the 26 log-target pairs it applies to, so the peek exists in the registered text and changes nothing in the result.

Two things the corpus does establish.. Two things survive the negative — a per-entity historical rate with no model and no attribute that reaches 6.1% to 109.3% of what the fitted model reaches above chance, and a validity check showing the generic target is not this paper’s published one. Both are in Appendix Appendix I.

12.3. From an explanation to a registered prediction

The paragraph above is an explanation written after seeing the corpus, and an explanation written after seeing the data is worth very little. This project had one opportunity to convert it into a prediction and took it.

What was registered, and when.. **PREDICTION.md** was committed *before* either held-out log was downloaded, let alone parsed: at the moment of that commit **data/normalized/** contained no parquet for either, **results/** contained no held-out result, and the script that downloads them did not exist. Two logs were held out — BPI Challenge 2011, a Dutch academic hospital, and BPI Challenge 2018, EU agricultural subsidies — both recorded in **PROTOCOL.md** §2 in the previous round as considered and not included, for size and parsing cost, and neither ever opened by this project.

The registered rule: R will be resolvably positive *iff* (i) $V(f \mid B_0)$ is resolvably positive and (ii) the opening field clears a threshold fixed numerically from the thirteen logs of Section 12. Two forms of (ii) are registered, and the reason is itself a finding.

The form we intended to register fails on its own training data.. Condition (ii) as first written — the *normalised entropy* of the opening field exceeds H^* — reaches an in-sample balanced accuracy of 0.600 on the nine pairs that have a denominator: four true positives, four false positives, one true negative. Fitted, $H^* = 0.331293$ amounts to “every log except Sepsis”. The flagship log has the second *lowest* normalised entropy of the nine, because dividing by $\ln k$ ranks a seven-valued field spread evenly above a fifty-seven-valued field with a heavy skew. We report that rather than quietly substituting a rule that works.

The form the data support.. Condition (ii) as $V(g \mid B_0) > V^* = 0.013302$ — the opening field’s own incremental AUC over the intake block — separates all nine training pairs: four true positives, five true negatives, balanced accuracy 1.000. It is also what the mechanism implies, since a field worth nothing over the intake block cannot absorb anything. **It was chosen after seeing that the entropy form failed, and the registration says so.** A rule selected on the training data can only be tested on data it was not selected on, which is what follows.

The test, and how little it establishes.. BPI Challenge 2011 is admitted by the registered rules and yields two scored pairs. On both, $V(f \mid B_0)$ is not resolvably positive — -0.0117 $[-0.0351, +0.0001]$ on duration and $+0.0006$ $[-0.0051, +0.0069]$ on handover — so both rules predict “not resolvable”, and on both the observed reduction is indeed not resolvable. Two predictions, two correct, zero errors for each rule. BPI Challenge 2018 is *excluded* on both targets by registered codes: `NO_TEST_VARIATION` on duration and `NO_HEADROOM` on handover, where the generic handover target has prevalence 1.0000 because a seven-valued resource stamp changes at least once in every one of 43,809 cases. Excluded pairs are reported with their code and are not scored; counting an exclusion rule as a correct prediction would be the cheapest possible way to win this test.

What that is and is not evidence for, said plainly.. Both registered rules survive: neither erred on a scored pair. **But both scored pairs are negatives,**

so the rules’ positive half was never tested at all, and the prospective test as executed shows only that the rules do not fire where the entity is worth nothing — which condition (i) alone guarantees. `PREDICTION.md` §4 stated before the logs were opened that at most four pairs could be scored and that the test therefore has very low power. It had less power than that. The precondition is registered, it is testable, and it has been tested once on two negatives; a reader who wants it established needs the next held-out positive, and `results/r42_new_since.csv` records that as of 2026-08-22 no event log has been published on 4TU since the cut-off, so this round leaves none unspent.

13. A Reporting Standard

We propose that a feature-value claim be reported as a surface over the arguments of Equation 1 rather than as a point, and that the minimum reportable form is:

1. **The baseline ladder.** Every already-recorded field admitted to B , named, with the admissibility criterion stated before it is applied, and V at each rung. A field the organisation already holds and does not admit is a decision, and its cost is the difference between two rungs.
2. **At least one integrating and one operating-point instrument.** On this data the two families differ by more than the effect. Reporting only one is reporting one of two answers without saying which question was asked.
3. **The operating range, not a chosen point.** If the claim is decision-analytic, the whole curve with intervals, and explicitly the region where the increment is not resolvable or is negative.
4. **The register’s population, if the feature is a lookup.** With the degradation regime named, because the regime moves the answer by more than the level does.

Figure 6 is that report for one instrument on this data, and Figure 5 is the same claim across instruments. Neither figure contains a number that is not in a result file, and every number in both is re-derived from those files by the checker described in Section 17.

13.1. The standard, as something a reader can run

A reporting standard that is a paragraph of prose makes it easier to ignore than to follow. We ship `fieldvalue`, a dependency-light Python package that computes the surface above and **declines, by design, to emit a single number**:

```
from fieldvalue import surface
s = surface(X, y, feature="ci_name",
            baselines={"intake": INTAKE,
                       "intake+group": INTAKE + ["group"]},
            metrics=["auc", "ap", "brier_skill", "nagelkerke"],
            thresholds=np.arange(0.05, 0.80, 0.025),
            population=[1.0, 0.75, 0.50, 0.25],
            order="opened_at", seed=20260819)
s.report()      # the minimum reportable form, as text
s.to_frame()    # the surface, tidy
s.spread()      # how much each axis moves the answer
s.plot()        # Figure 1
float(s)        # raises SingleNumberRefused, and says why
```

There is a command line, `python -m fieldvalue`, in which every argument that fixes one of the four axes must be named, because a tool that defaulted them silently would be the practice this paper is about.

The refusal is the standard, encoded.. `float(s)` raises and prints the range the surface actually occupies together with the three accessors that will answer a properly posed question. A caller who wants one cell must name it — `s.at(baseline=..., metric=..., threshold=..., population=...)` — which forces them to say which cell they are standing in.

An independent code path, and what it found.. The package's metrics are implemented in numpy from their definitions rather than delegated, so that re-deriving this paper's headline through it is a real check and not a tautology; `fieldvalue`'s test suite asserts agreement with scikit-learn to 10^{-12} , so the independence costs no correctness. `r46_tool_agreement.py` re-derives the surface of Section 7 through the package and compares it against Section 7's own pipeline: **20 of 20 quantities agree, the largest disagreement being 2.2×10^{-16}** . What is *not* independent, and the paper does not pretend it is: scikit-learn's `OneHotEncoder` and `LogisticRegression`, which both

paths call, and the cohort, which both read from the same loader. An error in the estimator or in the cohort would agree with itself.

The check earned its keep on its first run, and against this paper. Section 8’s claim that the reduction “runs 6.3% to 119.2% across the operating range” named an extremum that was not the extremum: over the 28 grid points where the denominator is resolvably positive it runs 4.7% at $\theta = 0.175$ to 134.1% at $\theta = 0.575$. Every literal in the old sentence was correct and the relation it asserted was not. It is the twelfth correction and it is in Section 17.

Tests, and a worked example on data that is not ours.. 91 tests, including the three propositions of Section 3.3 as test cases, so a proposition that stops holding breaks the library and not only the paper. `examples/worked_example.ipynb` runs the whole surface on UCI Adult, which is not one of this paper’s logs, and is the source of Section 4.5’s numbers.

14. The Question We Could Not Answer, Answered

The previous version of this paper had a section here titled “One Thing We Could Not Establish”. A third field, the knowledge-article reference, sits on the same `Open` row, is 100% populated, and costs nothing. Adding it to the baseline raises AUC to 0.805 and drives the measured value of item identity to -0.003 . Whether that figure belonged in the headline depended on whether the field is available at creation, and we could not determine it from the three files we held. That section named the missing evidence: an interaction detail file shipped in the same public collection, which we had not obtained.

We obtained it. `Detail_Interaction.csv` records 147,004 service-desk interactions.

What it settles.. 94,250 of those interactions — 64.1% of the file — never produce an incident at all, and 99.7% of those were resolved on the first call. Every one of them carries a knowledge reference, across 1,978 distinct articles. A field the incident process assigns cannot be populated on records the incident process never touches, so the reference is written by the service desk during the call. Of the incidents that do join an interaction, 92.4% of the cohort, the interaction opened before the incident in 100.00% of cases with a median gap of six minutes, and the desk’s recorded handling time on it had elapsed before the incident opened in 98.6%.

What it does not settle, and we say so.. Agreement with the closed record still cannot discriminate. On the 41,413 incidents whose interaction was worked to completion first, the knowledge reference agrees with its interaction’s value on 100.00% and the closure code — which is certainly not creation-time — on 98.97%. The reasoning of the previous version stands: an interaction is itself a closed record. The subset whose interaction had formally *closed* before the incident opened contains five incidents, and nothing is concluded from five.

The consequence, which removes our headline.. Admitting the reference carried by an interaction that was worked to completion before the incident opened — populated on 91.1% of the cohort — takes the item’s measured value from +0.103 to +0.001 $[-0.002, +0.003]$, a reduction of 99.7% against the intake-only baseline rather than 43.7%. Figure 7 draws the ladder.

Three nulls, because a result of that size is not believed on sight.. A collapse to zero could be collinearity — the reference being item identity relabelled — or dimensionality, 1,700 extra sparse columns at a fixed penalty being a burden the item must then overcome, or temporal coherence, since a knowledge article is used in bursts and a random grouping is not. **None survives**, and the three are constructed and measured in Appendix Appendix J: 78.8% of articles map to exactly one item against 80.7% for the opening group; five matched-mass random partitions of the same cardinality reach base AUC at most 0.6479 where the real field reaches 0.8041; and a partition matched on cell size *and* on each cell’s distribution over twenty time strata does not reproduce either number.

15. Threats to Validity, Each Measured

Every threat below has a measurement rather than a paragraph, and each measurement is somewhere in this paper already. They are collected here so a reader can see the whole list at once and check that none of them is answered with an argument.

Construct: the target is not what its name suggests.. Reassignment is routine handling, not error; it fires on 37.2% of test incidents. *Measured:* the ladder is reported at stricter target definitions in Section 6, and the paper says “reassignment” rather than “misrouting” throughout for that reason.

Construct: the free field is not the routing queue.. An earlier version asserted it was. *Measured:* three checks in Section 5 say otherwise — 50 distinct groups on the **Open** rows against 218 on the **Assignment** rows, one group holding 67.0% of openings but 18.4% of all activity, and a 15.1% match between the opening group and the first assignment. It is correction five.

Construct: the corpus does not measure this paper’s task.. The generic hand-over target is not the primary log’s own # **Reassignments** field. *Measured:* prevalence 0.927 against 0.411, agreeing on 46.0% of incidents. Section 12 states that the corpus measures a generic workflow outcome on twelve further organisations and not this paper’s task, which it would have been easy to write as though it did.

Internal: the split, and both censoring boundaries.. A random split on a process log leaks the future, so the split is temporal. *Measured:* 1,150 left-censored incidents are removed and are reassigned at 81.2% against 40.0% for those kept; re-measuring on cohorts truncated up to a month early moves the shrinkage between 42% and 45%, so the right-hand boundary does not drive it.

Internal: the estimator’s own knobs.. *Measured:* the L2 penalty is held fixed across every arm, and tuning it per arm on an inner split moves the headline reduction from 43.7% to 43.2%; **Priority** is a deterministic function of (**Impact**, **Urgency**) and dropping it moves intake AUC from 0.562 to 0.564. Two further estimator families are reported in Section 6.

Conclusion: the estimator might not estimate the estimand.. A ratio of two differences of two fitted AUCs is not obviously well behaved. *Measured:* against a world whose true reduction is known by exhaustive enumeration, the largest absolute bias is 0.0044 and bootstrap coverage runs 0.850 to 0.975 (Section 11). Where the denominator is not resolvably positive no ratio is printed at all.

Conclusion: the analysis pipeline might be wrong in one place.. Every number in this paper came from one pipeline until this round. *Measured:* an independent implementation re-derives the headline surface and agrees on 20 of 20 quantities to 2.2×10^{-16} , and found one error in this paper’s prose on its first run (Section 13.1). What the check cannot see is stated there: both paths share scikit-learn’s estimator and the same cohort loader.

External: one estate, and then three.. Measured: the whole surface is run on all three logs whose reduction is resolvable, and the baseline choice moves R by more than R on every one (Section 10). The population axis does *not* replicate, moving R by 0.085 on one log and 1.293 on another, and the paper says so.

External: the generality claim, registered and falsified.. Measured: 22 logs, 13 admitted, resolvable on three; `PROTOCOL.md` §8’s falsification condition is met and reported as met. The precondition that survives was then registered and tested prospectively, and produced two correct predictions, both negative, which tests only half of it (Section 12.3).

External: the audit reads only what is open access.. Measured: full text was retrieved for 369 of 600 sampled papers, 61.5%; the 231 that were not are recorded with their reason and are not coded. Nobody can show the loss is uncorrelated with the codes, and Section 16 does not assume it is.

The apparatus itself.. A verification harness that can be wrong about the paper is a threat to every number in it. *Measured:* eight holes have been found in this one and every one was found by its own corruption suite rather than by the harness or by reading; the suite carries 199 mutations, each of which changes the manuscript in one place and requires the checker to notice; the suite’s size, the checker’s coverage census and the purity guard that stops the checker modifying its subject are reported in Section 17 and in `REPRODUCE.md`.

16. Limitations

Era, decomposed into four measurements.. The primary log is six months of 2013 and 2014 from one estate, and practice has moved. “Practice has moved” is not one claim, however, and each of its four usual components has a proxy in data we hold. *(i) Discovery auto-populates the estate.* Ours is already 100% populated, so the measurement sits at the top of its own curve and better tooling cannot raise it; what a modern estate changes is which part of the register is populated, and Section 9 measures that. *(ii) Incidents arrive pre-stamped.* In 2014, 97.7% of incidents that came through the service desk already carried the item their originating call carried. A change that pre-stamps arrivals is a change from 97.7% to at most 100%. *(iii) Service mapping rather than item identity.* The service component — 272 values

in this cohort — reaches +0.078 over intake plus group against the item’s +0.103, so it captures 75.3% of it, and the item’s marginal over it is +0.023. (iv) *More intake channels, fewer central desks.* This is the largest sensitivity in the paper. Subsampling the cohort so that the largest opening group holds between 0% and 95% of it moves the reduction from 95.3% to 6.3%: when one desk opens everything the opening group carries almost no information and absorbs almost nothing, and when opening is distributed it absorbs nearly all of the item’s apparent value. A 2026 organisation with more channels sits at the end of that sweep where the free field matters *more*, not less. Two honest qualifications. Subsampling one estate’s opening groups varies the intake mix *within* one organisation, one tool and one period; it is a sensitivity, not a replication. What supports reading it as more than that is Section 12: the logs in the corpus whose opening field is nearly constant are exactly the ones that show no effect, so the sweep’s prediction is confirmed on data it was not fitted to.

Data quality. Section 9 replaces the caveat this paragraph used to make. What remains is that we measure population and not *accuracy*: a register that is fully populated with wrong values is not modelled here, and we have no way to detect one in a public export.

Free text, and a search rather than a regret. Neither of the published logs carries a usable short description, and every real deployment has one. Kapel et al. [11] rank two free-text fields above CI Name for a related target. We therefore searched: every attribute of every one of the 22 parsed logs, against a criterion declared in `PROTOCOL.md` §7 before the search ran. Of 596 attributes, zero satisfy it. The field a referee will name, UCI 498’s `u_symptom`, is 3.4% unique with a mean length of 10.9 characters: a coded symptom, not a description. Two amendments were needed to stop the criterion admitting formatted identifiers, and they are recorded. The threat is real, it is the most important open question this study leaves, and it cannot be tested on public process-mining data. We report the search rather than the regret.

Construct and mechanism. Reassignment is a proxy for misrouting and also fires on legitimate escalation; the within-item shuffle preserves item membership but not other correlates of identity, so it bounds the proxy share rather than isolating it; and the opening group’s distribution drifts between our halves. After round sixteen’s withdrawals, we do not claim a measured direction for the mechanism.

The corpus measures a different target from the rest of the paper.. Section 12 applies a generic handover target so that one rule can run on twenty-two logs. On the primary log that target and the log’s own reassignment count agree on 46.0% of incidents. The corpus therefore bounds the generality of the *estimand and the practice*, not of this paper’s particular magnitude, and we say so there rather than letting thirteen log names imply thirteen replications.

What is still outside reach.. No deployment, no organisational partner and no practitioner validation; a single author with no institutional affiliation; public benchmark data whose newest log is from 2019. None of the four is fixable by further analysis and we would rather state them than have them inferred.

The operational statement is a detection statement.. Net benefit counts detections against reviews at a stated exchange rate. It does not say that a detected incident is a corrected one, that a desk given the warning would route differently, or what a correct routing is worth. Section 8 also reports a region of thresholds in which the item is worth nothing or less than nothing over the group-aware baseline, and we have no mechanism for that beyond over-confident predictions from a high-dimensional model.

17. Corrections, and the Classes They Fall Into

Twelve errors of our own are reported as results rather than edited away. A paper whose thesis is that unexamined choices set the answer cannot suppress its own, and that is the whole of the reason: it is not candour and it is not a confession, it is the same evidence the rest of the paper is made of, turned on the paper.

The twelve incidents, each with its cause and the section it changed, are in Appendix Appendix B and in `REFeree-LOG.md`. What is worth the main text is not the incidents but the **classes**, because a class is transferable and an incident is not. Five account for all twelve.

Class 1 — a null drawn at the wrong level (three instances).. A control built to bound a mechanism was constructed at row level while the quantity it bounds lives at item level, so the association it was meant to destroy was destroyed by construction and the null could only return zero. The same control, rebuilt at the right level, was then swept only to 800 cells while

the leg it bounds uses 2,929, and the margin was read at the cell count that maximised it. A third instance ran the control on one of the two rungs whose difference it bounds. **The general form: a null is a claim about a level, and a null evaluated at a different level from its target is not a weak test but a vacuous one.** This is the class this project has repeated most, and the reason it repeats is that a vacuous null returns a comfortable number rather than an error.

Class 2 — a quantity printed without an interval (one instance)... An operational factor was published as a point estimate, on an unmatched pair of models, in a paper that elsewhere declines to resolve +0.0017 past two decimals. **The general form: a paper’s precision standard has to apply to its headline, and the headline is exactly where the pressure not to apply it comes from.**

Class 3 — an asserted meaning that no measurement supports (four instances)... A central field was described as the routing queue for several versions when it is the group that logged the incident; a public dataset was described as lacking a field it has, on every case, while a script in our own repository was already reading it; an estimator-free asymmetry turned out to be an algebraic identity between two entropies sitting one division apart in the same file; and a factor’s sign turned out to be a tie-break inside a single tied block holding 93.1% of what the baseline nominates. **The general form: a sentence about what a number *means* is a claim, and it needs a measurement of its own. The apparatus that checks values never looks at it.**

Class 4 — a relation between correct numbers (three instances)... An extremum named at the worst of the points we happened to have tabulated rather than at the extremum; a count of a set reported as the length of an interval; and, found this round by the independent code path of Section 13.1, a range across the operating range that was the range across five named thresholds. In all three, **every individual literal was correct and sitting in the correct place**, and every one passed a checker that re-derives each literal from a result file. What was wrong was the word joining them — *reaching, in a contiguous run from, across the operating range.* **The general form: a directional or quantifying word is a claim with no number of its own, and a verifier built around literals is structurally unable to see it.** The checker now compares each named extremum against the

extremum and each stated count against the run length; that closes three instances and not the class, and the corruption suite carries mutations that break a relation while leaving every value correct.

Class 5 — an admission decision left unexamined (one instance).. The previous version reported that it could not establish when a field is written, named the file that would settle it, and did not obtain the file — which sits in the same public collection as the three it did obtain. Obtaining it took the headline from +0.103 to +0.001. **The general form: the cost of not looking is invisible, so it is never weighed against the cost of looking.** This paper’s own thesis predicts the failure, which is why it is reported as a correction and not as progress.

Three further classes, on the apparatus rather than on the paper.. The verification harness has had eight holes and *every one was found by the corruption suite, never by the harness and never by reading.* Three of the classes generalise past this repository. *A consumer placed above its producers:* a coverage census and a guard lint each consumed a list that later code appended to, so both silently stopped seeing the checks written below them; moving the block is not a fix for a hole whose cause is order, and both are now functions called from one late block, with a lint that reads the checker’s own source and fails if a check is registered after them. *A harness that can modify its subject:* a check that read the corruption suite’s size by importing it *ran* it, and every corruption reported caught because the nested checker refused to start; a clean result was produced by an apparatus that had checked nothing. The checker now records the manuscript’s SHA-256 before it reads anything and asserts it is unchanged before it reports. *A checker that reads a pipeline’s output cannot see a pipeline that never runs:* one analysis script had not parsed for thirteen rounds while 936 checks passed against the files it was supposed to produce; the reproduction script now parses every script before it starts.

The direction, which is the only summary worth drawing.. Eight of the twelve flattered the result, one excused the paper’s principal limitation, two would have made the result larger had they been noticed earlier, and one removed the result altogether. We draw the obvious inference about the direction in which unchecked assertions travel, and we do not claim the thirteenth will be different.

18. Conclusion

What a recorded field is worth is not a number. It is $V(f \mid B, m, \theta)$, and on this decision every one of those arguments moves the answer by more than the effect we set out to report.

Vary the baseline: knowing which configuration item an incident concerns is worth +0.183 AUC against four intake fields, +0.103 once one further field the organisation already records is admitted, and +0.001 $[-0.002, +0.003]$ once a second is. Vary the metric on those same scores: the reduction runs 43.7% to 60.3% over six instruments, and the figure we previously published is the smallest of them. Vary the operating point: it runs 4.7% to 134.1%, and in a band above the base rate the item is resolvably harmful. Vary the register's population and the regime by which it empties: at half population the item is worth +0.082 or +0.064 depending on whether the tail or the core is missing. Vary how the organisation takes work in: across intake mixes the reduction runs 6.3% to 95.3%.

None of those variations is a defect in any instrument or an error in any measurement. Each is correct about the question it asks. The point is that a single number answers one of those questions without saying which, and that the person writing the business case is the person choosing.

We are asked what a buyer should conclude, and on the scope this paper can speak to the answer is plain. On this task, this estate and this target, a register that costs money to build and maintain adds nothing measurable over two fields the organisation already records, and a business case built on +0.183 is pricing that register together with both of them. We do not say that about registers in general; Section 12 is why we cannot, and Section 16 lists what would have to be true for the sentence to travel.

We tried to make the finding general and it did not go. A pre-registered protocol over 22 public logs admits 13 and finds a resolvable reduction on three; the registered generality claim is falsified and Section 12 reports it. What the corpus does establish is the precondition: the effect needs an entity that predicts the outcome at all, and on most public process logs there is not one. What we offer in its place is Section 13: a minimum reportable form for an incremental-value claim, demonstrated on data where following it removes the paper's own headline. That is the strongest evidence we can give that it is worth following.

Acknowledgements and Reproducibility

The manuscript was subjected to seventeen rounds of adversarial review, the later rounds machine-assisted, in which each round was asked to break the previous version’s claims; the corrections in Section 17 came from that process. The protocol of Section 12 was registered in the repository before the analyses it governs were run, in a commit that adds no result file. The author set every research question, adjudicated every proposed correction against the data, and is responsible for the content. The datasets are public [8, 22]; code, figures and a checker that recomputes every quantity here: <https://github.com/sudhirdixit1/cmdb-field-admission>.

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Appendix A. The Proposition Constructions

Each construction below is executed by `scripts/r41_propositions.py`, which asserts the resulting numbers against the claim and exits non-zero if

any of them stops holding. A referee who distrusts a proof can run it; a referee who distrusts the code can read the closed forms.

Proposition 1

Every model's score takes one of two values, so a model is the pair of counts (a, b) : positives and negatives in the high block. With P positives, $N = \rho P$ negatives, $\alpha = a/P$, $\beta = b/N$ and $\pi = 1/(1 + \rho)$,

$$\text{AUC}(\alpha, \beta) - \frac{1}{2} = \frac{1}{2}(\alpha - \beta), \quad \text{AP}(\alpha, \beta) - \pi = \alpha \left(\frac{\alpha}{\alpha + \beta\rho} - \pi \right).$$

The first is half Youden's J and follows from the midrank definition of AUC with ties taking half credit. The second follows from evaluating the precision-recall steps at the single threshold that separates the two blocks.

Both baselines are the fully tied model $(\alpha, \beta) = (1, 1)$, which has $\text{AUC} = \frac{1}{2}$ and $\text{AP} = \pi$, so $V = m(\text{model}) - m(\text{baseline})$ under both metrics. Choose the naive leg $(\alpha_1, \beta_1) = (d, 0)$ and the honest leg $(\alpha_3, \alpha_3 - d)$ for $\alpha_3 \in [d, 1]$. Then $\alpha - \beta = d$ on both legs, the AUC increments are equal, and

$$R_{\text{AUC}} = 0 \quad \text{exactly,} \quad R_{\text{AP}}(\alpha_3) = 1 - \frac{g(\alpha_3)}{g(d)}, \quad g(\alpha) = \alpha \left(\frac{\alpha}{\alpha + (\alpha - d)\rho} - \pi \right).$$

g is continuous and decreasing on $[d, 1]$, so R_{AP} sweeps $[0, 1 - g(1)/g(d)]$, and the upper end tends to 1 as ρ grows.

Why it must be built in integers.. The first implementation passed proportions and rounded a and b , so $\alpha - \beta$ was only *approximately* equal between the legs and R_{AUC} came out at 4.4×10^{-4} rather than at zero. A proposition claiming an exact zero has to be constructed in integers: ρ is an integer and dP and dN are integers, so for integer $a_3 \in [dP, P]$ the matching $b_3 = a_3\rho - dN$ is an integer and the equality is exact. ρ is part of the construction and not a fixed constant: near $\alpha = d$ the AP increment is steep, so at $\rho = 199$ a single integer step of a_3 moves R by 0.038 and $r = 0.05$ cannot be reached, while at $\rho = 9$ the steps near zero are 0.0007 and the reachable maximum is only 0.818. The declared family sweeps seven (P, ρ) pairs, each kept under 1.2 million rows so that a referee can run it, and the chosen cell is recorded per target in `results/r41_prop1.csv`.

Proposition 2

The witness applies $\varphi(p) = p^{0.55}$, strictly increasing on $(0, 1)$, to the scores of one leg of a four-model ladder built on 20,000 synthetic rows. The rank vectors before and after are asserted identical. Under ROC AUC and average precision R moves by exactly 0.000000; under Brier skill, Nagelkerke R^2 and net benefit at $\theta = 0.20$ and $\theta = 0.40$ it moves by 2.200, 1.645, 4.255 and 2.681.

Proposition 3

Eighteen synthetic estates are generated by varying three declared knobs: how tightly the opening field is coupled to the entity (0.15, 0.55, 0.85), how much of the entity’s effect on the outcome the opening field carries (0.15, 0.50, 0.80), and how strong the entity’s effect is (1.2, 2.2). Nothing in the generator is fitted, so it cannot encode the answer. For each estate the spread of R along each of the four axes is computed with the other three held at declared defaults. For each ordered pair (X, Y) the search reports the two estates whose X -spreads agree to within 5% of that axis’s observed range and whose Y -spreads differ by the most; the pair is constructible if that difference reaches 25% of the Y axis’s range.

Both tests are *relative* to the axis’s own range. An earlier version used a fixed absolute tolerance of 0.02, which made every pair with $X = \text{threshold}$ unconstructible for a reason that is about units — the metric axis moves R by tenths and the threshold axis by whole multiples — and not about the proposition. The full search, including the two pairs that do not reach the criterion and the gaps they do reach, is in `results/r41_prop3.csv`.

Appendix B. The Twelve Corrections, One by One

Each entry names what was wrong, what replaced it, and the section it changed. Section 17 groups them into classes; this appendix is the record, and `REFeree-LOG.md` carries the objection each one answers.

1. **A null that could not fail.** The mechanism’s floor was originally drawn by assigning cell labels per *row*, so two incidents sharing an item landed in different cells and the association was destroyed by construction. The null could only return zero, and we published a margin of 89 points against it (Section Appendix D).

2. **The same null, rebuilt, was still a knob.** Rebuilt at item level it was swept only to 800 cells while the leg it bounds uses 2,929, and the reported margin was taken at the cell count that maximised it. At matched granularity the margin is 3.5 points at $z = 0.9$ and we now withdraw it. This is the fifth time a null in this project has been drawn at the wrong level.
3. **An asymmetry that was an algebraic identity.** The estimator-free evidence for the direction of the mechanism was a ratio of two uncertainty coefficients, which is identically the ratio of two marginal entropies. It was published for eight rounds with both entropies sitting in the same results file, one division away.
4. **A factor printed without an interval.** The capacity-table factor was first given as a point estimate, on an unmatched pair of models, in a paper that elsewhere refuses to resolve +0.0017 past two decimals.
5. **A central field whose meaning we asserted.** The opening group was described as the routing queue for several versions. It is the group that logged the incident (Section 5).
6. **A claim about a public dataset that our own repository refuted.** We wrote that the third public ITSM log has no affected-item field. It has one, populated on every case, and a script in our own repository had been treating it as such (Section Appendix G). This was the sole support for describing the study as single-organisation of necessity, which we had presented as a constraint rather than a choice. It was a choice.
7. **An operational factor whose sign was a tie-break.** Correction four gave that factor an interval; this one asked what it measured. At a 5% review capacity 93.1% of what the naive baseline nominates is drawn from one tied block, and reordering rows inside that block — which changes nothing any model knows — moves the naive arm from -26 to $+608$. We withdraw the factor and rebuild Section 8 on net benefit, which never breaks a tie. The replacement number is 1.07, not 4.3 (Section Appendix F).
8. **A control run on one of the two rungs it bounds.** The shuffled-item encoder null was run on the group-aware rung only, while the reduction it is meant to bound is computed from two rungs, and its residual under boosting is eleven standard errors from zero rather than noise. Run on both rungs, the correction moves the boosting reduction by 3.4 percentage points, and upward (Section 6).

9. **An extremum that was the worst of the points we had named.** We wrote that the group-aware increment turns negative, “reaching -16.1 $[-23.0, -8.9]$ per thousand at $p_t = 0.50$ ”. The word *reaching* names an extremum. The extremum is -21.1 $[-27.7, -14.4]$ at $\theta = 0.525$, which is 31% larger; 0.50 was in the list of five thresholds our own table happened to name and 0.525 was not. Every literal in that sentence was correct and was computed from a file already in the repository (Section Appendix K).
10. **A count and a run reported as the same set.** We wrote that the increment is resolvably positive “at 20 points, in a contiguous run from 0.100 to 0.425”. Fourteen are in that run; six sit at 0.675 and above. Counting a set and describing an interval are different operations and this sentence performed one and reported the other (Section Appendix K).
11. **An open question we left open with the evidence one download away.** The previous version reported that it could not establish whether the knowledge-article reference is available at incident creation, named the file that would settle where the field comes from, and did not obtain it. The file is in the same public collection as the three we used. Obtaining it shows the reference is written by the service desk on calls that never become incidents at all, and admitting it takes our headline from $+0.103$ to $+0.001$ (Section 14). We record this as a correction rather than as progress because the paper’s own thesis is that an unexamined admission decision sets the answer, and we left ours unexamined for a round.
12. **A range across the operating range that was a range across five named points.** We wrote that under net benefit the reduction “runs 6.3% to 119.2% across the operating range”. Those are the values at $\theta = 0.325$ and $\theta = 0.500$, two of the five thresholds our own table happened to name. Over the 28 grid points where the denominator is resolvably positive the reduction runs 4.7% at $\theta = 0.175$ to 134.1% at $\theta = 0.575$. Every literal was correct and the phrase *across the operating range* was not, which makes this the third instance of the class corrections nine and ten belong to. It was found by the independent code path of Section 13.1, on that path’s first run, which is the only reason it was found at all: the checker had a check on each of 6.3 and 119.2 and no check on the word *across*.

Grouping	levels	AUC	gain
CI Type (aff)	13	0.671	+0.027
CI Subtype (aff)	61	0.657	+0.013
Service Component WBS (aff)	256	0.722	+0.078
CI Name (aff)	2,554	0.748	+0.103
CI Name, marginal over the service component	—	—	+0.023

Table C.6: At what resolution configuration data carries its value, over a baseline that already contains the opening group. Levels are counted in the training half.

Appendix C. Which Layer of Configuration Data Pays?

The field-admission question of Section 6 has a companion that is more directly a buying decision, and it is where this paper’s novel finding sits: not whether configuration data is worth anything, but at what resolution, and whether what pays is the configuration *attributes* at all. Four nested groupings of the same estate are all 100% populated and free once the item is known. Measured over the intake-plus-group baseline of 0.644:

A 256-way service-component grouping captures three quarters of what instance-level identity is worth, and instance identity adds +0.023 over it. The classification layers proper — CI type and subtype — carry little, and CI Subtype scores below CI Type despite being finer, which is what a fixed penalty does to a sparser column. This is the same +0.023 that Section Appendix D reports for admitting the service component to the baseline, seen from the other side: the two are the same quantity, and a reader who regards an application or service register as something an organisation has without a configuration-management practice should read the CMDB’s marginal contribution here as that figure.

Why these fields are in the ladder and not in the baseline.. The obvious question is why CI Type and CI Subtype, being 100% populated and free, are not simply admitted to the CMDB-free baseline alongside the opening group. The answer is the second admissibility criterion of Section 5 and it needs no model: both are exact functions of item identity. Neither varies on a single one of the 2,929 items, so both exist only where the item register exists, and admitting them would be admitting the CMDB to the CMDB-free baseline. The service component is the case this argument does not dispose of, and the quantitative gap is instructive: it varies on 58 of 2,929 items, carrying 8.7%

of incidents, where the opening group varies on 565 items carrying 92.5%. The line between the two fields is real and it is a matter of degree, which is why we report the headline both ways rather than choosing.

The signal is an outcome history keyed on identity.. A per-item reassignment rate computed on the training half and applied to the test half as a lookup table — no model, no other field, no configuration attribute of any kind — scores 0.744, against 0.745 for item identity alone under the full estimator and 0.748 for the paper’s complete model. Whatever the CMDB supplies here, it is not the item’s attributes; it is a stable identifier under which six months of outcomes can be accumulated. The defensible reading is that maintaining such an identifier across a changing estate — dedupe, reconciliation, retirement — is itself a configuration-management function, and we think that reading is right. But it is a different claim from “the CMDB is worth +0.103”, and an organisation that can key its ticket history on any stable token can have most of this without one.

Estate concentration.. Required scope is bounded by a property of the estate needing no model: the top 8 items generate 30.2% of training incidents, the top 64 70.5%, and the top 128 — 5.0% of the training vocabulary — 82.0%, all from a frequency count before funding anything. Identifying only those items recovers most of the value. Averaged over five split points, the top 8 recover 56% of the +0.103, the top 64 recover 88% and the top 128 recover 93%; across those splits the three range over 52–58%, 81–93% and 90–96%. These are min–max spreads over a design choice, not bootstrap intervals, and we write them as ranges to keep the distinction. We give the mean and range rather than one split’s curve because a single curve is not monotone at this resolution: at $k = 32$ the across-split spread is 9 points. The same holds with the group removed from the baseline — 89% at $k = 64$ — so this is a property of the estate’s concentration, not of our field-admission decision. Taken with the resolution ladder, the practical reading is that the marginal AUC of the 2,801 items outside the top 128 is small, and that is where the cost of a CMDB programme lives.

Appendix D. Where the Difference Goes

The two rows of Table 2 differ by 0.080, which is not itself informative: for any two variables, the drop in one’s gain when the other joins the baseline equals the drop in the other’s. Read four ways, the item is worth +0.183 over

intake and +0.103 once the group is present; the group is worth +0.082 over intake and +0.002 once the item is present.

This section is weaker than the version we published previously, and the weakening is a finding rather than a concession. Of the four legs that supported the overlap reading, one has been withdrawn as an algebraic identity, one no longer resolves once its null is matched properly, and two survive.

What survives, first.. Adding the opening group to a model that already knows the item is worth +0.0017 [+0.0001, +0.0034]. It clears a matched-dimension null of -0.0009 ± 0.0005 that no draw reaches, so it is not the cost of adding columns; but across split points and penalties it ranges +0.0002 to +0.0072, so the honest statement is that its unique contribution is *under* 0.01 *AUC* and *not resolvable more finely*. Because the real column is nearly constant within an item while the null’s is drawn per row, this null is matched on dimension but not on that collinearity, and a sceptical reader should treat the interval as the weaker of the two claims.

What survives, second.. Item identity does not determine the group, but predicts it far better than chance: a modal-group lookup on item identity reproduces which group logged the incident on 90.3% of test incidents against 78.6% for always guessing the largest, while class-balanced accuracy reaches only 34.1%. Item identity resolves the coarse contrast — central desk or specialist team — and much less of the fine one. The two variables share 1.47 bits of mutual information.

A margin that was a knob.. Randomising the group label *within* each item — which destroys the label’s own identity while preserving what the item implies about it — retains 91% of the group’s +0.082 gain. We previously compared that against a floor built by randomising within a random partition *of items* at 49 cells, matching the opening group’s own cardinality, which retains 42%, and reported the 49-point margin as evidence that item identity specifically was doing the work.

That comparison is not matched. The real leg partitions on item identity, which is 2,929 cells; the floor was drawn at 49. Retention rises monotonically with the fineness of the partition, so the margin is a function of a knob that was set to the value maximising it. The stated ground for stopping the sweep — that at one cell per item the null becomes the real thing — is wrong, because a *random* partition of n items into n cells is not the identity partition and stays routing-blind. Swept out to matched granularity the floor

retains $87.5\% \pm 3.6\%$ against the real leg's $91.0\% \pm 1.6\%$: a margin of 3.5 points at $z = 0.9$, against this project's own resolvability bar of $|z| > 3$. We therefore withdraw the margin. What the sweep shows is that retention is high at fine granularity whether or not the cells mean anything, which is a fact about granularity and not about routing.

A withdrawn measurement: the asymmetry was an identity.. We previously reported that, without any estimator, item identity carries 60.4% of the opening group's information while the group carries 19.6% of the item's, and read the asymmetry as evidence that the mechanism runs from item to group. The reading is unsupported and the measurement is circular. The uncertainty coefficient $U(A | B) = I(A; B)/H(A)$, and mutual information is symmetric, so the ratio of the two coefficients is identically $H(\text{item})/H(\text{group})$ — a function of two marginal entropies and of nothing else. On this log that ratio is 3.09, and the two coefficients, their shuffled floors, and the floor-corrected excesses we published all stand in that same ratio to six decimal places. Any two variables whose entropies stand in that ratio produce those two numbers, whether or not either determines the other. We withdraw the asymmetry and the directional claim it supported. The mutual information itself, 1.47 bits, is a real measurement and is reported above; its *direction* is not measurable this way.

The reduction is graded in the field's resolution.. Coarsening the group and re-running the ladder gives reductions of 27%, 28%, 36% and 44% at two, four, eleven and 49 levels. This is a consistency check rather than an independent prediction, because at each resolution the reduction is close to that coarsened field's own gain over intake — by the identity above, the same statement as the first measurement — and the two coarsest points have overlapping intervals, so only the trend is claimed. A single binary flag — central desk or not — already recovers 61% of the group's baseline gain and 63% of the reduction. The field that must be admitted is close to one bit, far cheaper to reproduce elsewhere than a 49-way taxonomy.

Is the field really free? One objection goes to the heart of the reading. The opening group may be free only because a human at the desk already knew what the ticket was about — in which case it is undocumented configuration knowledge, and a baseline containing it is not a CMDDB-free baseline. A third reading is at least as likely and we had missed it: the contrast may be an intake-channel indicator, separating an out-of-hours or overflow or automated

route from the central desk, in which case it is neither independent of the item nor configuration knowledge held by a person. The one-bit result makes the question concrete without settling it: the contrast doing most of the work is the central desk against everything else, and Section 5 declines to say what everything else is. The measurement is the same under all three readings; the moral is not. If the group is independent, a CMDB is worth half what the naive comparison says. If it is tacit configuration knowledge, our $+0.103$ is a lower bound and the organisation already holds much of what the CMDB would tell it, unpaid for. The log cannot identify which is true, so we choose none of them. What survives either way is that the business case cannot be written against the intake block alone.

What we do not claim.. The reverse framing — that the item column “stands in for” the group — is not established. We measured it, by randomising items within groups, and obtained 44% of the item’s gain — but a routing-blind equal-size partition of items retains *more*, 56%, and one whose cell sizes follow the group’s own mass profile retains 25%. The leg is floor-dominated and we exclude it. Nor do we claim a direction of causation: the log records no counterfactual, so whether the item drives who logs the ticket or both follow a third cause is not identified here. Our claim is about what an analyst’s baseline choice does to a measured number, which does not depend on which is true.

Two fields we did not admit, and why.. Our baseline is not a terminus either. Hour of day and day of week are free and creation-time; adding both moves the item’s value from $+0.103$ to $+0.099$. **Service Component WBS (aff)** is harder: 100% populated, free, and admitting it takes the measured value to $+0.023$. We exclude it as near-deterministic in the item — only 58 of 2,929 items carry more than one value — so it is configuration data under another name. That criterion cuts both ways: 2,060 of 2,554 training items also map to a single opening group, though those items carry just 8.8% of incidents, so the high-volume ones are the multi-group ones. We know of no principled threshold that admits the group and excludes the service component. A reader who draws the line differently should read our headline as $+0.023$ — which is the decision this paper is about, and why we report a ladder rather than a number.

Appendix E. Why the Instruments Disagree

Three mechanisms are testable and we test all three. Each makes a prediction that can fail, and the first one did.

Aggregation, which is the largest.. AUC is $\int \text{TPR} d\text{FPR}$, so the increment a feature buys can be attributed to regions of the operating range without any modelling choice: the integral is additive over a partition of the FPR axis and the parts sum to the whole, which we assert rather than assume. We predicted the increment would be concentrated somewhere the net-benefit grid does not reach. It is not. Over the 31-point grid the group-aware model runs at every false-positive rate from 0.009 to 0.997, and the item’s advantage is *diffuse*: the largest of ten equal FPR bands carries 18.2% of it and three bands are needed to reach half. AUC sums that whole profile; net benefit at one θ reads one point of it. The two are not disagreeing about a fact.

Baseline degeneracy, which explains where net benefit is lowest.. At $\theta = 0.200, 0.300$ and 0.325 the intake block acts on every incident in the test half: as a decision rule it *is* treat-all, and the group-aware baseline acts on 0.955 to 0.978 of them, barely distinguishable from it. An increment measured against a rule that does nothing comes out nearly the same on both runs, and their ratio near one. Calling a baseline degenerate when it acts on more than 95% or fewer than 5% of arrivals — a threshold we declare before applying it — the reduction runs 0.047 to 0.435 across the 11 grid points where both baselines are degenerate and 0.581 to 1.168 across the 5 where neither is.

Calibration, which is real and provably not the whole story.. The item-aware model has calibration slope 1.040 and the intake block 1.391. Recalibrating on a held-out tail of the training half moves the net-benefit reduction by up to 0.253 where the denominator is safe and the proper-score reductions by up to 0.014. It moves the AUC and average precision reductions by exactly zero, because a monotone rescaling cannot change a rank statistic — a prediction that would have failed loudly had the recalibration not been monotone, and did not.

And a tie convention the two rank statistics do not share.. The intake block emits 23 distinct scores for 13,637 incidents. Average precision’s default value equals its *negatives-first* bound, to machine precision, on all four models,

Tie-breaking policy	naive extra	honest extra	factor
random — the only implementable one	+271	+63	4.3
oracle: positives first in each tie	−26	+24	—
adversarial: negatives first	+608	+104	5.8

Table F.7: Extra reassignment-bound incidents surfaced at a 5% review capacity by adding item identity, against a baseline that includes the opening group (honest) and one that omits it (naive), under three policies for ordering rows *inside* a tied block. The oracle and adversarial policies use the outcome and are unavailable to an analyst; together they bracket what tie-breaking alone can do. Medians of 400 paired bootstrap draws over test rows.

because the precision-recall curve is evaluated where a tied block has been admitted whole; AUC’s default equals a random tie-break to within 0.0004. Put both on the only convention a desk could implement, the reduction is 43.6% under AUC and 71.2% under average precision. The gap widens rather than closes, which is the outcome we did not expect and report because we did not.

Appendix F. Why the Capacity Framing Was Withdrawn

The paper already disclosed that the naive baseline cannot rank finely: four low-cardinality intake fields, one of them the redundant Priority column, emit only 23 distinct scores over 13,637 test incidents, so at 5% capacity just 47 rows sit strictly above the cut and the rest come from a tied block of 1,944. It said the factor was therefore partly about rank resolution rather than information, that the two had not been separated, and moved on. Separating them destroys the number.

The naive top-5% is a draw from a tie.. Of the 682 incidents the naive baseline nominates, 635 — 93.1% of the review budget — come from a single tied block, and that block is reassigned at 0.534 against 0.372 overall. How many of the right incidents the draw yields is therefore fixed by the block’s own base rate and by nothing the model knows. The 47 rows the model does rank strictly above its cut contain 9 reassignment-bound incidents, a rate below the base rate: at the top of its own ranking the naive baseline is worse than guessing. The group-aware baseline is in a different position, with 590 of 682 ranked strictly above the cut and only 92 drawn from a block of 246.

The factor’s sign is set by the tie-break.. Table F.7 prices that. Ordering rows inside the tied block optimally — which uses the outcome, so no analyst can — reverses the sign of the naive contrast: item identity surfaces 26 fewer incidents than the intake block alone, not 271 more. Ordering them adversarially gives +608. Nothing any model knows changes between those three rows. A quantity whose sign is set by how a coin lands inside a 1,944-row block is not a measurement of information, and we withdraw it.

A reader may object that the oracle policy favours the coarser arm by construction: it has the larger block to reorder, so it has the most to gain from an ordering it cannot obtain. That is the objection restated as the finding. A contrast whose magnitude is governed by the size of one arm’s tie block, rather than by what either arm knows, is measuring the block. The random policy is the only implementable one and its value is the one an analyst would see; but its value is the expected yield of a draw from a block whose composition is a property of the cohort, not of the ranking, and the paper should not have read a difference in information off it.

The obvious repair is impossible.. It is natural to answer the objection by giving the naive baseline a finer representation, and we tried: cross-fitted target encoding of the intake block, per field and as a single composite key, on the same machinery used for the item column in Section 6. It cannot work. The four intake fields take 23 distinct combinations, so every row sharing a combination shares a score under *any* function of those fields, and combinations unseen in training collapse onto the training prior — the composite encoding emits 19 distinct scores, fewer than the one-hot model’s, not more. The tie block is not an artifact of our estimator. It is what having four low-cardinality intake fields is.

What survives is narrower and we state it as such: the honest arm’s extra catches are positive under every tie policy, including the oracle (+24 [+9, +40]), so knowing the affected item does surface more reassignment-bound incidents at this capacity than the group-aware baseline does. What does not survive is the ratio, and with it the claim that omitting the free field overstates the operational gain by more than it overstates the AUC gain.

Appendix G. The Second Organisation

An earlier version described this study as single-organisation of necessity, on the ground that of the three public ITSM incident logs in common use, one

records the affected item on every incident, the second on 0.2%, and the third had no such field at all. The third clause was false, and our own repository contained the evidence: the BPI Challenge 2013 incident log [22] from Volvo IT, recorded in VINST, carries a `product` attribute with 704 distinct values, exactly one per trace, on all 7,554 traces and none missing. It is a coarser identifier than an instance-level CI name, but it is an affected-item field, and the challenge’s own headline question was ping-pong behaviour between teams — the same outcome our primary target measures.

We therefore run the ladder there too. The item analogue is `product`; the free field is the `org:group` on the first event, of which there are 338; the intake block is `impact`, the one intake-form field the two logs share. The target is at least one change of `org:group` along the case, which fires on 50.6% of cases, with a stricter variant at two or more changes (27.2%). We use the same temporal 70/30 split and the same estimator.

One difference cuts against us and we state it rather than let a reader find it. On BPI Challenge 2014 the opening group matches the incident’s first `Assignment` group only 15.1% of the time: it is the desk that logged the ticket, a different object from the routing sequence that defines the target. On BPI Challenge 2013 the opening group *is* the first term of the `org:group` sequence whose variation defines ping-pong. The coupling is therefore strictly tighter there, the free field is correspondingly worth more (+0.188 against +0.082), and the Volvo reduction should be read as an upper bound rather than as a second draw from the same distribution.

Appendix H. The Corpus, and What the Rules Exclude

Twenty-four files across seven domains, fetched by DOI with a recorded SHA-256 for each; 22 logs parse. The registered rules admit 13 across six domains and exclude nine with a code. Two exclusions are worth stating as findings rather than as bookkeeping. In BPI Challenge 2012 and in all five BPI Challenge 2020 sub-logs, *every case is opened by the same actor*: the opening resource stamp has cardinality one, there is no free opening field to admit, and the question this paper asks cannot arise. In the Hospital Billing log the only reusable entity, `diagnosis`, is missing on 65% of first events and no other attribute reaches the registered cardinality floor.

Appendix I. Two Things the Corpus Does Establish

The condition that governs, and it is measured on the primary log.. The exclusions are not bookkeeping; they are the answer. BPI Challenge 2015’s five municipalities are the closest thing the public corpus has to a controlled replication — one process, five organisations — four are admitted, and none shows a resolvable reduction. Their opening resource stamp has cardinality 7 to 18. BPI Challenge 2012 and all five BPI Challenge 2020 sub-logs have cardinality *one*, which is why the protocol excludes them. A nearly constant opening field cannot absorb anything, and Section 16’s intake-mix sweep measures precisely that dependence on the primary log: as the largest opening group’s share of arrivals goes from 0% to 95%, the reduction falls from 95.3% to 6.3%. A sweep run on one organisation predicts which logs in the corpus will show the effect and which will not, and that is the strongest thing the corpus produced.

A precondition, not a moderator.. We attempted the prediction we set ourselves in advance: regress the reduction on log properties measurable before any model is fitted — entity cardinality, its concentration, the mutual information between f and g , trace length, whether g is a person or a team, and four more. On nine points with eleven predictors the leave-one-out R^2 is -1.323 against a permutation null of 2,000 shuffles at $p = 0.448$: the fit predicts a held-out reduction worse than its own mean does. We report that and show no in-sample fit in its place. What the corpus does show is coarser and more useful: the effect requires an entity that predicts the outcome at all, and on most public logs it does not.

A generic target is not the published target, and we checked.. The protocol’s handover target counts distinct opening groups in a trace. The primary log ships its own reassignment count, which is what this paper’s other sections use. They are not the same construction and they do not agree: the generic target fires on 92.7% of incidents against the published target’s 41.1%, and the two agree on 46.0% of them, barely above what independence between two marginals of that size would give. The tautology control of `PROTOCOL.md` §4.3 splits with them — under the generic target the largest opening group has the higher rate, under the published target the lower. The corpus measures a generic workflow outcome; it does not measure this paper’s task on twelve further organisations, and it would have been easy to write as though it did.

Appendix J. Three Nulls Behind the Collapse

Section 14 reports that admitting the knowledge reference takes item identity from +0.103 to +0.001. A result of that size is not believed on sight, and these are the three nulls that stand behind it.

A collapse to zero could be collinearity — the reference being item identity relabelled — or dimensionality, 1,700 extra sparse columns at fixed penalty being a burden the item must then overcome, or temporal coherence, since a knowledge article is used in bursts and a random grouping is not. None survives.

Collinearity. 78.8% of knowledge articles map to exactly one item, against 80.7% for the opening group, which absorbs less than half of the item’s value rather than all of it. Determinism at that level does not collapse a marginal.

Dimensionality. Five matched-mass random partitions of the same cardinality — built by slicing a permutation rather than by `searchsorted` on cumulative mass, which is the bug that killed correction three — reach base AUC at most 0.6479 and leave the item worth +0.094 to +0.099. The real field reaches 0.8041 and leaves it worth +0.001.

Temporal coherence. Five further partitions matched on cell size *and* on each cell’s distribution over twenty equal-count time strata, so that every synthetic cell has the same size and the same profile through time as the real one it copies. They reach base AUC at most 0.6483 — the real field is 0.1558 above that — and leave the item worth +0.092 to +0.101. We did not build cells from contiguous blocks of the time-ordered cohort, which would give perfect temporal coherence and would be useless: the split is temporal, so every test cell would be unseen in training and the null could not fail. A null that cannot fail is this project’s most-repeated defect and we did not add a sixth.

One thing this ladder does *not* do is condition on a subset. The knowledge reference enters as a feature column on the unchanged cohort: incidents whose interaction does not qualify carry an explicit missing token and stay in the model, the split and the test set, so every rung is measured on the same 13,637 test incidents.

What this does to the paper.. Our thesis is that the measured value of a field is set by an admission decision the analyst makes silently. The strongest available demonstration of that thesis is that it removes our own headline, and this is that demonstration. We do not claim the knowledge reference is

proved creation-time; we claim the balance of evidence has moved decisively, that a paper whose headline is contingent on excluding a field must report what admitting it does, and that +0.103 should be read as the value of item identity *given that one particular free field is admitted and another is not*. Section 13 is the general form of that sentence.

Appendix K. The Decision Curve, in Full

Vickers and Elkin [26] supply the instrument the question actually needs. Fix a threshold probability p_t — the risk at which a desk judges a review worth doing — act on every incident scored at or above it, and compute

$$\text{NB}(p_t) = \frac{\text{TP}}{n} - \frac{\text{FP}}{n} \cdot \frac{p_t}{1 - p_t}.$$

The weight $p_t/(1-p_t)$ is the exchange rate the threshold implies: at $p_t = 0.25$ one missed reassignment-bound incident is worth three needless reviews. Net benefit is in true positives per incident, so we report $1000 \times \text{NB}$, which reads as reassignment-bound incidents caught per thousand arrivals net of the reviews spent catching them. It assumes no capacity, it is defined across the whole range of exchange rates a reader might hold rather than at one we picked, and — the point that matters after Section Appendix F — a threshold admits or excludes a whole tied block, so no tie is ever broken.

The scores are probabilities enough for this.. Net benefit is meaningless for a model whose scores are not calibrated, and a one-hot logistic model with 2,554 sparse columns at a fixed penalty is not obviously calibrated. Brier scores run 0.232 for the intake block, 0.206 with the opening group and 0.189 with the item; calibration slopes run 1.391, 1.185 and 1.040, so the intake block is the poorly calibrated arm and the item-aware model is close to ideal. Because a slope away from one moves net benefit, we repeat the whole analysis on scores recalibrated on a held-out tail of the training half — the model refitted on the first 85% of training, scored on the last 15%, and the scaling applied to test, with no test outcome entering it. No conclusion below changes.

Where the item pays, and by how much less than it appears to.. Over a grid of 31 thresholds from 0.05 to 0.80, the group-aware increment is positive with its interval excluding zero at 20 points. Those 20 are not one run: 14 of them

p_t	over intake + group	over intake alone	ratio
0.20	+17.2 [14.0, 20.4]	+19.3 [16.0, 22.5]	1.12
0.30	+64.9 [59.8, 70.2]	+69.9 [64.7, 74.8]	1.08
0.40	+40.7 [33.0, 48.9]	+97.2 [88.6, 106.0]	2.39
0.50	-16.1 [-23.0, -8.9]	+84.1 [75.5, 93.2]	—
0.60	+1.8 [-3.7, 7.2]	+39.8 [32.5, 47.2]	—

Table K.8: What item identity adds, as net benefit per 1,000 arrivals, under the two baselines. A ratio is not quoted where the group-aware increment’s interval includes zero or lies below it, because a ratio whose denominator is crossing zero is not a quantity.

form a contiguous block from 0.100 to 0.425, the region either side of this cohort’s base rate, and the remaining 6 sit at 0.675 and above, on the far side of the negative band described below. An earlier version of this sentence said all 20 lay in the run; every number in it was correct and the relation asserted between them was not (Section 17). The naive increment is positive and resolved at 29 of the 31. At the threshold where item identity is worth most, $p_t = 0.325$, it adds 86.9 reassignment-bound incidents per thousand arrivals over the group-aware baseline and 92.8 over the intake block: omitting the free field overstates its value there by a factor of 1.07. This is the number that replaces the withdrawn 4.3, and it is smaller than the ratio of 1.8 between the same two gains measured as AUC.

And where it does not pay at all. Above the contiguous region the increment decays and then turns negative: at four grid points between 0.475 and 0.575 the group-aware increment’s interval lies entirely below zero, and its minimum over the whole grid is $-21.1 [-27.7, -14.4]$ per thousand at $\theta = 0.525$. An earlier version gave the minimum as $-16.1 [-23.0, -8.9]$ at $\theta = 0.50$, which is the worst point among the five this paper had chosen to name rather than the worst point on the grid; the understatement was 31% and it flattered us (Section 17). At a one-for-one exchange rate, adding item identity to a model that already knows the opening group makes the desk worse off on this task. Locating that band operationally matters, and the direction is the uncomfortable one: a *higher* threshold means acting on *fewer* arrivals, so the harmful band is where a capacity-constrained desk sits, not where an unconstrained one does. Across the four thresholds the group-aware item model acts on 17% to 37% of arrivals. We have no mechanism to offer for the sign beyond the obvious one — a model with 2,554 item indicators

can be confidently wrong where a coarse model abstains — and we report it because it is the honest reason the capacity framing produced the number it did. The earlier version measured the overstatement at the one operating point where its own group-aware arm had the least to give.

One dataset, one feature: the admissibility reduction R in every cell

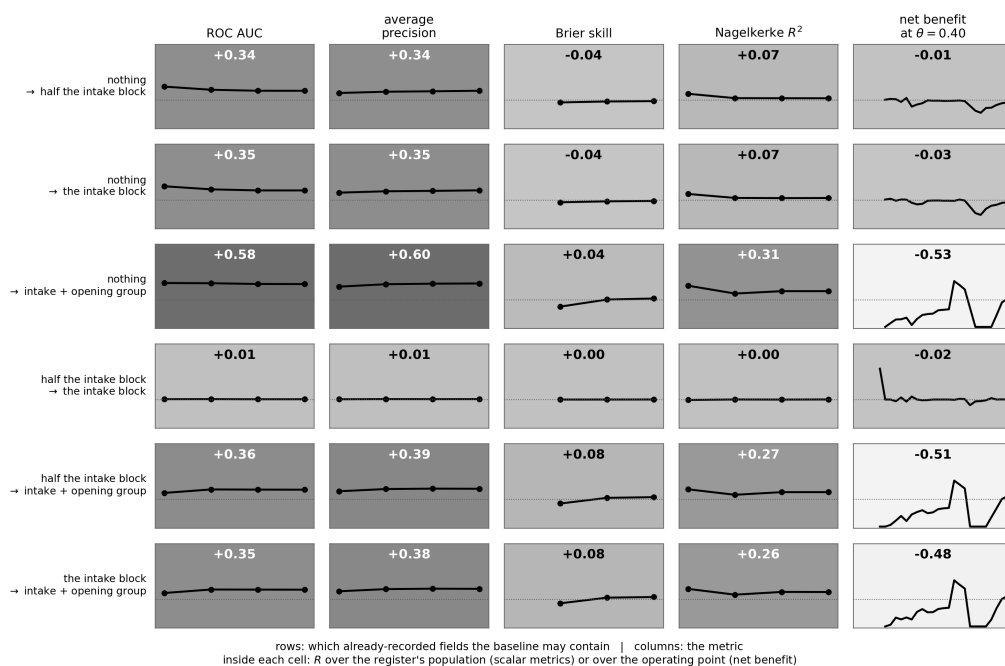


Figure 1: The same data, the same feature, and every defensible answer at once. Each cell is the admissibility reduction R for item identity on BPI Challenge 2014: rows are which already-recorded fields the baseline is allowed to contain, columns are the metric, and the curve inside a cell runs over the register's population (scalar metrics) or over the operating point (net benefit). The reference values printed in the cells run -0.53 to +0.60. **WHAT THIS FIGURE DOES NOT SHOW:** it carries no intervals — every cell is a point estimate, and the bootstrap intervals are in Table 3 and in `results/r44_spread.csv`; it shows one log, and the same grid for two further organisations is in the same file; and it shows the reduction, not the increment, so a cell where the entity is worth almost nothing to begin with looks the same as one where it is worth a great deal. The in-cell curves are clipped to $[-1, 2]$ and the unclipped values are in `results/r47_cells.csv`.

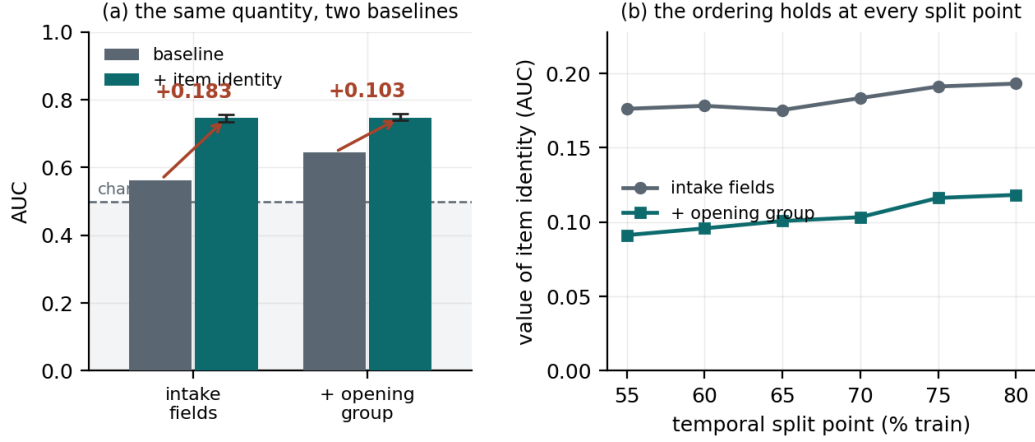


Figure 2: (a) The same quantity under two baselines, on the metric’s full range; the whisker on each treatment bar is the paired bootstrap interval of that bar’s own gain. (b) The ordering holds at every temporal split point from 55% to 80%.

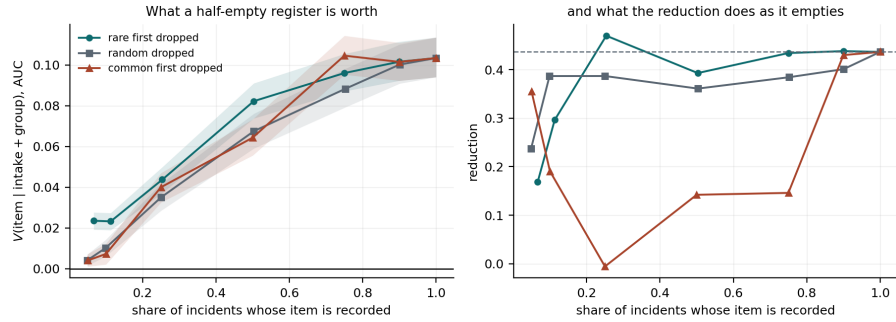


Figure 3: What a partially populated register is worth. Left: the item’s value over intake plus the opening group, against the share of incidents whose item is recorded, under three degradation regimes. Right: the reduction of Equation 2 on the same axis. The estate is concentrated enough that 246 of 2,929 items carry 90% of the incidents, which is why dropping the tail costs so little.

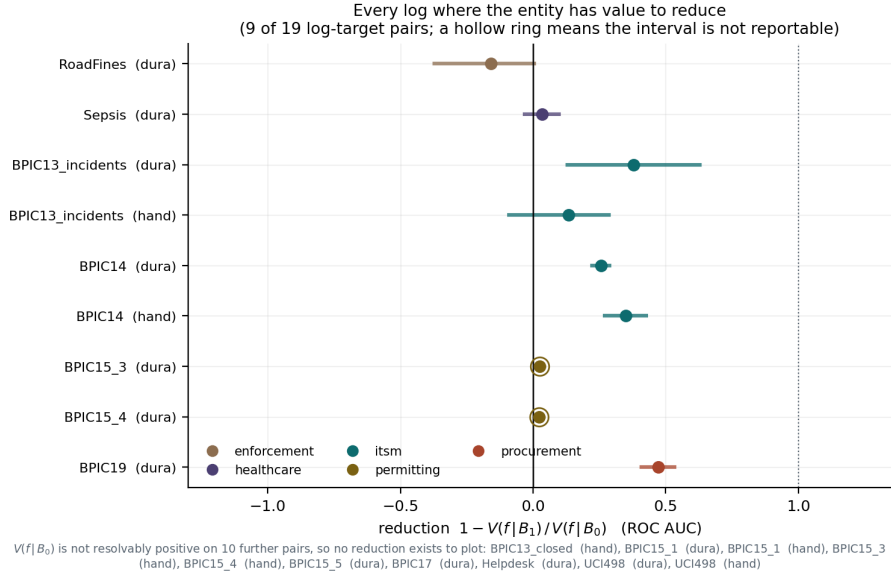


Figure 4: Every log-target pair where the entity is resolvably worth something over the intake block, so that a reduction exists. A hollow ring marks a pair whose reduction has no reportable interval because the denominator is not positive in every draw.

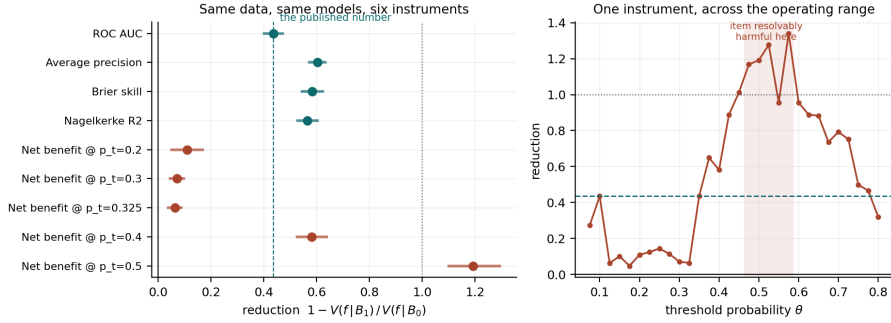


Figure 5: Left: the reduction under six instruments, paired bootstrap intervals, the published AUC figure dashed. Right: the same reduction under one instrument across the operating range. The shaded band is where the item's increment over a group-aware baseline is resolvably negative.

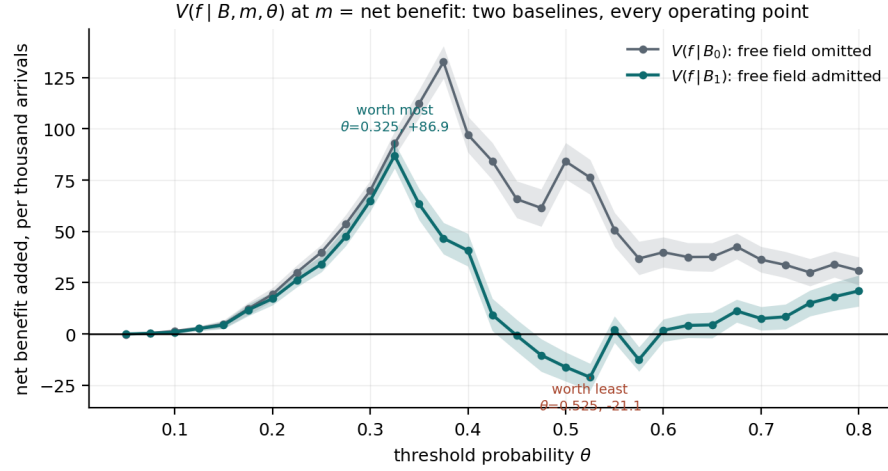


Figure 6: $V(f | B, m, \theta)$ at $m = \text{net benefit}$, both baselines, every operating point, with paired bootstrap bands. A single number for “what the item is worth” is a choice of one x -coordinate and one of these two curves.

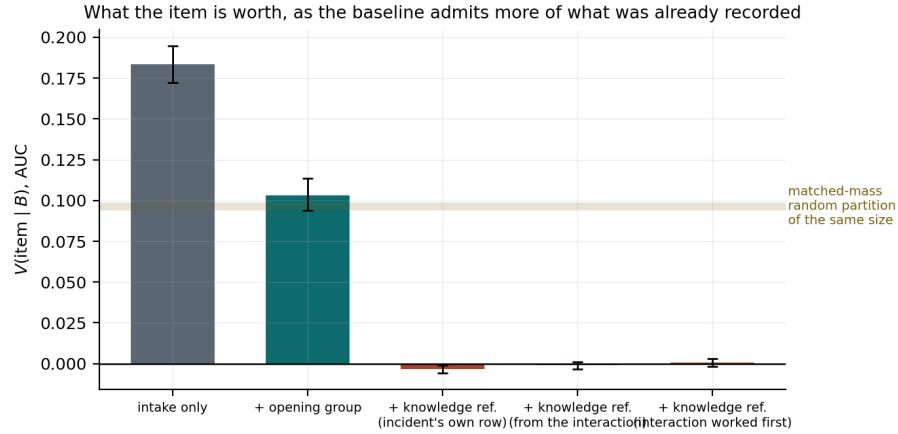


Figure 7: What item identity is worth as the baseline admits more of what the organisation already recorded. The shaded band is the range left by five matched-mass random partitions of the same cardinality as the knowledge reference.

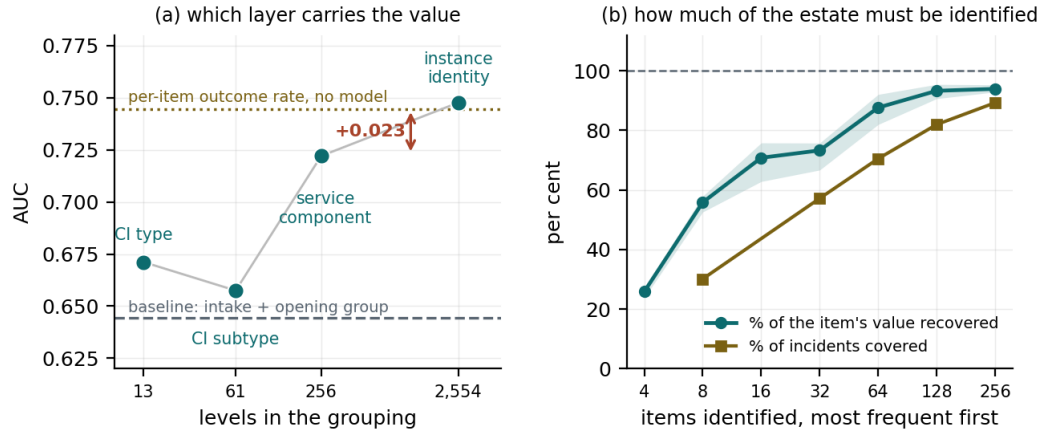


Figure C.8: (a) Which layer carries the value, over a baseline that already contains the opening group. The dotted line is a per-item outcome rate applied as a lookup, with no model and no configuration attribute. (b) How much of the estate must be identified: the share of the item's measured value recovered by identifying only the k most frequent items, against the share of incidents those items account for. Band: min-max across five split points.

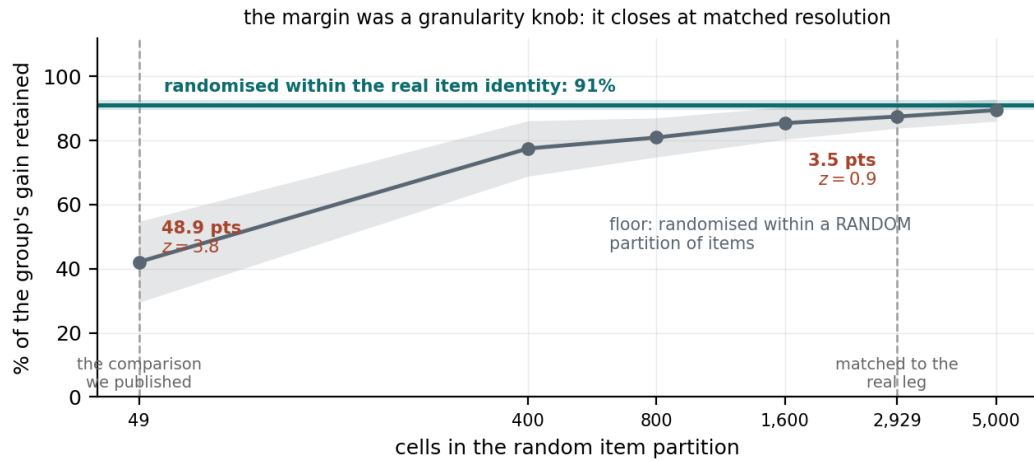


Figure D.9: Randomising the group label within each item, against the same randomisation within a random partition of items, as a function of how fine the partition is. The margin we published was measured at 49 cells; the leg it bounds partitions on 2,929. It closes as the partition approaches the granularity of the real leg.

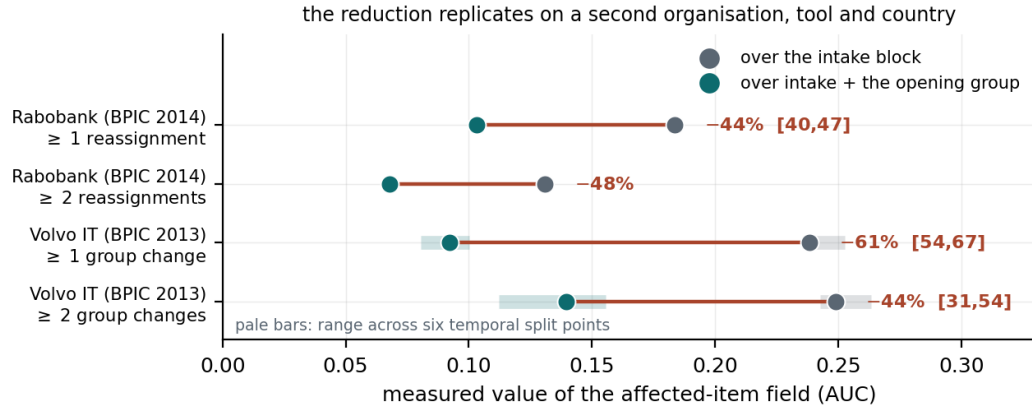


Figure G.10: The ladder on both organisations, at two target thresholds each. Each segment runs from the item's value over the intake block to its value once the opening group is admitted; the label is the reduction, with its paired bootstrap interval where one was computed. Pale bars are the min-max range across six temporal split points, which was computed for the Volvo rungs only.

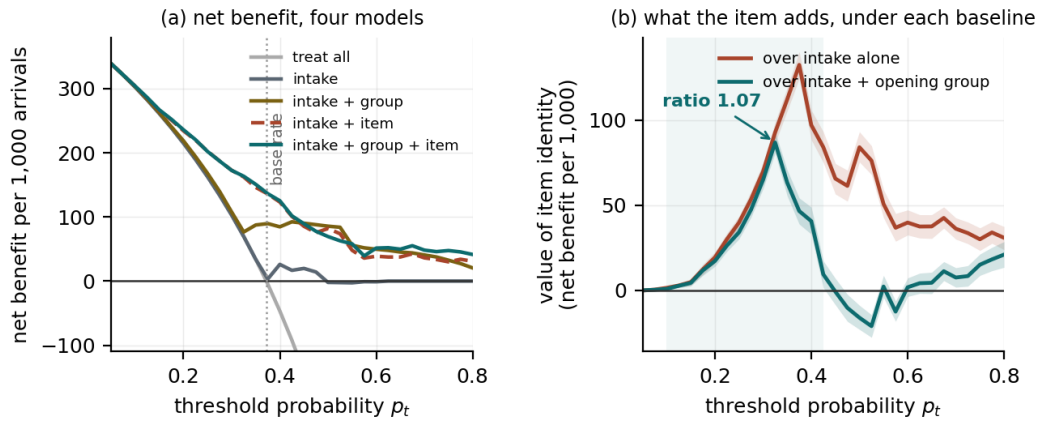


Figure K.11: (a) Net benefit per 1,000 arrivals for the four models and for treating every arrival, across threshold probabilities. (b) What item identity adds under each baseline, with 2,000-draw paired bootstrap bands. The shaded region is where the group-aware increment's interval excludes zero. The two curves are indistinguishable until the threshold passes this cohort's base rate.